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Advances in Hazard Management for Machine Learning based Safety-Critical Systems

Operating in Unpredictable Environments

by

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Thesis for the degree of Doctor of Philosophy

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University of Southampton

Abstract

Faculty of Engineering and Physical Sciences

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Doctor of Philosophy

Appendix F Detailed Theory of AIC Systems Approach

by

Haider M. Al-Shareefy

This appendix accompanies the doctoral thesis entitled *Advances in Hazard Management for Machine Learning based Safety-Critical Systems Operating in Unpredictable Environments*.

Appendix F provides a comprehensive technical reference for the AIC Systems Approach introduced in Chapters 3 and 4, presenting the full methodological detail underlying each stage of the approach in a format intended to support practitioner implementation, auditor review, and future research extension.

The appendix opens with the Predictive Complicatedness Resolution Matrices (§F.1), documenting the Deep AIC-Matrix and Actions Matrix as the foundational tools for exhaustive pairwise interaction enumeration within a defined complexity field. These matrices operationalise bounded lateral thinking by constraining the Architect's analytical attention across all interaction categories within the AIC ontology, providing the structural basis for systematic tail exploration. The General Complex Ontology Definition (§F.2) formalises the type system underlying the approach, defining ComplexType, OrgXPTYPE, PurposeClarity, and Outcome enumerations, establishing the semantic vocabulary from which all AIC interaction characterisations are constructed and against which all predictions are classified.

Collectively, Appendix F constitutes the practitioner-facing technical reference for the end-to-end AIC Systems Approach, providing the implementation depth that complements the conceptual and case study presentations in Chapters 3, 4, and 5.

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Research Thesis: Declaration of Authorship

Print name: Haider Al-Shareefy

Title of thesis: Advances in Hazard Management for Machine Learning based Safety-Critical Systems
Operating in unpredictable environments

I declare that this thesis and the work presented in it are my own and has been generated by me as the result of my own original research.

I confirm that:

This work was done wholly or mainly while in candidature for a research degree at this University. Furthermore,

Where any part of this thesis has previously been submitted for a degree or any other qualification at this University or any other institution, this has been clearly stated. Furthermore,

Where I have consulted the published work of others, this is always clearly attributed. Furthermore,

Where I have quoted from the work of others, the source is always given. With the exception of such quotations, this thesis is entirely my own work. Furthermore,

I have acknowledged all main sources of help. Furthermore,

Where the thesis is based on work done by myself jointly with others, I have made clear exactly what was done by others and what I have contributed myself. Furthermore,

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..... Date: 16/03/2025

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Appendix F Detailed Theory of AIC Systems Approach

F.1 Predictive Complicatedness Resolution Matrices

F.1.1 The Deep ArcMatrix (Deep AIC-Matrix)¹

Knowing that an intelligent drone is trespassing into a train track zone is not enough information to make a sound engineering judgement (prediction) on how to deal with it, because the nature of intelligence opens up a can of possible reasons why it is doing so. One of the most critical risks of intelligence is “intention”. Suppose we deal with regular non-learning complexes, such as track zone electric power lines. In that case, they have one goal: to remain securely attached to the infrastructure to ensure the delivery of electrical power. Learning complexes, such as intelligent drones, can adapt their goal settings depending on their Scenario. Therefore, knowing only that there are access train tracks is not enough and more assumptions need to be clarified. Here is where the AIC-Matrix come in to solve.

Key Components of the Deep AIC-Matrix:

1. Rows (Source Complexes):

- Represent the entities initiating interactions or actions in the complex. These may include intelligent agents (e.g., drones) or non-intelligent systems (e.g., power lines).

2. Columns (Sink Complexes):

- Represent the entities receiving the effects of the interactions. Sink complexes may include operational zones, physical infrastructure, or other agents in the environment.

3. Cell Details (Interaction Emergence Decomposition): Each cell in the matrix captures the interaction between a specific source and sink complex, decomposed into the following elements:

- **i. Supra Source:** Describes the encompassing supra complex (larger complex) from which the source complex emerges or operates. This provides a view of the source's operational context.
- **ii. Supra Source PrimeP:** Details the encompassing **PrimeP** (primary purpose) of the source complex within its supra source. This clarifies the higher-order intent driving the source's behaviour.

¹ The tool used in section 6.6.2.

- **iii. Source Goal:** Identifies whether the source complex has a defined goal related to the sink complex. If no goal exists, it is explicitly noted as "no-goal."
- **iv. Source Goal Type:** Categorises the goal as either an **AIC-type goal** (appreciative, influence, or control) or "unknown" if the goal type cannot be classified.
- **v. Source Action:** Specifies whether the source complex takes an action toward the sink complex. If no action exists, it is noted as "no-action."
- **vi. Source Action Type:** Categorises the action as an **AIC-type action** (appreciative, influence, or control) or "unknown" if the action type is unclear.
- **vii. Source Action Impact Type:** Evaluate the impact of the action on the sink complex:
 - **Supportive:** The action positively contributes to the sink complex's goals or operations.
 - **Obstructive:** The action creates challenges or hinders the sink complex.
 - **Neutral:** The action has no significant effect.

Below is a general representation of Deep AIC-Matrix:

Table F.1 General format of Deep AIC-Matrix

	Sink complex A	Sink complex B
Source complex A		Supra Source: encompassing Supra Complex. PrimeP: encompassing Supra Complex PrimeP Goal: goal, or no-goal Goal type: AIC or unknown. Action: action or no-action. Action type: AIC or unknown. Effect: Obstructive, Supportive, Neutral.
Source complex B	Supra Source: encompassing Supra Complex. PrimeP: encompassing Supra Complex PrimeP Goal: goal, or no-goal Goal type: AIC or unknown. Action: action or no-action. Action type: AIC or unknown. Effect: Obstructive, Supportive, Neutral.	

Table F.1 introduces the **Deep AIC ArcMatrix (AIC-Matrix)**, a framework designed to decompose and analyse the complicated interactions among intelligent and non-intelligent systems within an operational context. It is particularly relevant for learning complexes Scenarios, where their adaptability and potential intentions must be understood to ensure effective engineering judgments and complex safety. The AIC-Matrix provides a structured mechanism for evaluating the relationships between **source complexes** (entities initiating actions or goals) and **sink complexes** (entities receiving the effects of these actions or goals). Each matrix cell details the

nature of the interaction between a source and a sink by decomposing it into well-defined aspects. For example, the following is an example application of the AIC-Matrix:

Table F.2 Eagle Robot case study example of applying a Deep AIC-Matrix to describe the relationship between train track zone and anonymous drone

	anonymous drone	train track zone
anonymous drone		Supra Source: Adversarial Scheme. PrimeP: Disrupt Train Network operations. Goal: Strike passing train. Goal type: Control. Action: Intrude. Action type: Control. Effect: Obstructive.
train track zone	Supra Source: Train Network. PrimeP: Safely transport people and goods. Goal: manage drone access. Goal type: Appreciation. Action: no-action. Action type: Appreciation ² . Effect: Supportive.	

Table F.2 illustrates an application of the Deep AIC-Matrix to analyse the interaction between an anonymous drone and the train track zone within the Eagle Robot case study. The matrix captures the relationship between these two entities, with each cell detailing the interaction's goals, actions, and effects. The anonymous drone is identified as part of an Adversarial Scheme with a Prime Purpose (PrimeP) of disrupting train network operations. Its goal is to strike a passing train, categorised as a Control-type goal, executed through an intrusion action of the control type. This action is obstructive and clearly threatens the train network. In contrast, the train track zone is part of the Train Network Supra Source, with a PrimeP to ensure the safe transportation of people and goods. Its goal is to manage drone access, which is classified as an appreciative-type goal because it requires no direct action and relies instead on appreciative oversight. This inaction supports the drone's obstructive behaviour, which does not align with the train track zone's overarching purpose of safety and operational continuity. This example demonstrates the utility of the AIC-Matrix in breaking down and understanding complicated interactions between intelligent agents and non-intelligent systems.

The case study does not explicitly define the drone's purpose in the example. Our assumption that the drone's purpose is adversarial is based on its obstructive impact on the safety of the train track zone for passing trains. However, it's essential to justify such assumptions in practice. Constructing the AIC-Matrix requires a Chain-of-Thought approach to approach information from the Actions Matrix to arrive at the answers in the AIC-Matrix. Therefore, resolving the AIC-Matrix

² A no-action is an appreciative type behaviour, since no-action entails no influence or control, and place the system in an appreciative situation.

requires predicting the drone's adversarial nature. Additionally, the zone does not influence or control drone action; it only allows it to occur because no action is taken. Considering the PrimeP, we can deduce a goal to manage drone access, which is an appreciative goal due to a lack of capability. If the current goal is to appreciate, then inaction that results in an inability to influence or control the drone's behaviour is also appreciative. Consequently, the output impact of the train track zone's inaction behaviour supports autonomous drone intrusion.

Note how we elaborated a detailed and comprehensive Scenario out of a single action word, "permit access." If we read the Scenario, knowing AIC intuition, we can validate the reasoning behind the assumption. While resolving the complexity using AIC-Matrix, we made predictions about the observed Scenario. Hence, AIC-Matrix adds an artefact that demonstrates sound articulation of the problem and explains the decisions made.

AIC Factorisation Technique

Applying AIC factorisation within a Chain-of-Thought framework provides a pragmatic approach for dissecting the emergence of complex interactions into patterns of causally contributing Scenarios. This approach not only assists in unpacking current complex capabilities but also aids in predicting and preparing for potential future challenges by exploring various shifts in perspective. The approach involves five steps: identifying the source and sink complexes, determining the action undertaken by the source on the sink, categorising the action as appreciative, influencing, or controlling, defining the likely purpose behind the source's behaviour as appreciation, influence, or control, and assessing the impact of the source's action on the sink's primary purpose, whether supportive, obstructive, or neutral.

Table F.3 AIC emergence factorisation approach

Step 1) Predictive query: What is the interaction's source and sink complexes? Guiding Prompt: Review the observed Scenario and identify a source and sink complex. For example, Source: Eagle Drone. Sink: GPS module.
Step 2) Predictive query: Given the source complex, what is the larger Supra Source and PrimeP? Guiding Prompt: Define the Supra Source complex of which the source is part. For example, Supra Source: Policing Force. PrimeP: Ensure the safety and security of the Train Network.
Step 3) Predictive query: What action is taken by the source upon the sink? Guiding Prompt: Identify an action in the format of a very in the present tense that is being undertaken by the observed source. For example, Action: Synchs

Step 4) Predictive query: What is the AIC type of the action? Guiding Prompt: Specify the type of actions (Appreciative, Influencing or Controlling) Example, Action type: Controlling
Step 5) Predictive query: What is the most likely source's goal behind the source behaviour and its AIC type? Guiding Prompt: Given the context of the observed interactions, define the most likely source's goal and its AIC type. For example, Goal: Geolocate its position within the track zone area. Goal Type: Control.
Step 6) Predictive query: How does the source complex's action affect the sink's behaviour? Guiding Prompt: Define the most likely impact the identified action has on the sink's chances of manifesting its primary purpose. Example, Impact type: Neutral.
Source: An affecting complex whose behaviour is the source of interaction. Sink: An affected complex by the behaviour of the source. Supra Source: A larger encompassing complex where the source is part of. PrimeP: Source complex primary purpose. Action: Action being taken by the source. Action type: Appreciative, Influencing or Controlling. Goal: Source's intended end Scenario that is expected to happen. Goal type: Appreciation, Influence or Control. Impact type: Source action impact on sink.

F.1.2 Actions Matrix

When facing a Confusingly Complicated Scenario, the ArcPM may find it helpful to construct an initial, simplistic appreciation of the observation. Such appreciation allows the architect to consider a bird's-eye view (holistic) picture of what is being observed, abstracting out the details. The choice of complexes to be considered in the initial assessment is objective, based on the work done before their construction. For example, we introduce the superficial matrix in Predictive Thinking Pipeline 2 in the first stage, after objectively identifying the interacting complex in Predictive Thinking Pipeline 1. The matrix is constructed by mapping all elements of the concerning complexes to each other, such that each complex is mapped to all complexes in the observed whole. The main components of the Actions Matrix construct are:

- a Source complexes (represented in matrix rows).
- b Sink complexes (represented in matrix columns).

c Source Actions in every cell.

The following table represents a generic format of Actions Matrix:

Table F.4 General Format of Actions Matrix

	Sink complex A	Sink complex B
Source complex A		Source complex A Action
Source complex B	Source complex B Action	

Table F.4 presents a generic format for the Actions Matrix, a quick tool for assessing the predictability of complexity fields by analysing the specific actions taken by source complexes in relation to sink complexes. The rows represent the source complexes initiating the actions, while the columns represent the sink complexes that receive these actions. Each cell in the matrix captures the specific action undertaken by the source complex towards the sink complex.

For example, suppose the following problem Scenario: An anonymous Drone intruded into a bounded train-track zone. To construct an Actions Matrix, first, we define the observed complexes of concern set: {anonymous drone, train track zone}. An action matrix can be constructed as follows:

Table F.5 Example Actions Matrix

	anonymous drone	train track zone
anonymous drone		Intrude
train track zone	Permit access	

Table F.5 illustrates an Actions Matrix applied to a Scenario involving an anonymous drone and a bounded train track zone. The Scenario begins with identifying the observed complexes of concern: {anonymous drone, train track zone}. The Actions Matrix is then constructed to capture the specific interactions between these complexes. As a source complex, the anonymous drone executes the action of intrusion into the train track zone, reflecting a direct and deliberate engagement with the complex. Conversely, the train track zone, as a source complex, acts as permitting access, indicating a passive response that allows the drone to continue its intrusion. This matrix effectively captures the dynamics of the Scenario by isolating the actions of each complex and demonstrating their interaction.

Note that the initial description of the problem did not explicitly expose what the train track zone is doing to the intruding drone. Here, we had to think harder to predict or assert what action track zone is performing towards the drone. Such aspects remain invisible from the explicit accurate description as a tacit assumption. The Actions Matrix forced us to consider carefully what kind of action the train track zone is doing to the drone. Therefore, the Actions Matrix tell us that our initial

perception of the problem was incomplete. However, there is more invisible knowledge about “drone intrudes into the zone, and zone permit access of drone”, which we need to make explicit and visible. Here is where the approach of constructing an AIC-Matrix comes in handy.

The following is an example taken from our AVOIDDS case study, which shows how AIC actions matrix looks like:

Table F.6 Example AIC Actions Matrix for AVOIDDS problem domain. Red is unresolved influence, Blue is control, Green is appreciation, Black is resolved influence.

									
		+capture	+capture	+assure	+assure	+capture	+predict	+capture	+validate
	-visually complicate		visually complicate	visually complicate	- operationally complicate	+control	-reduce reliability	visually complicate	- complicate
	-visually complicate	operationally complicate		visually complicate	operationally complicate	operationally complicate	-reduce reliability	- operationally complicate	visually complicate
	+predict	+predict	predict		+assure	+predict	+optimise	-predict	+validate
	+inform	+monitor	monitor	+inform		+inform	+monitor	+control	+inform
	-visually complicate	+expose	visually complicate	-visually complicate	-visually complicate		-reduce reliability	visually complicate	visually complicate
	+predict	compute visual complexity	+compute visual complexity	+assure	+assure	+compute visual complexity		+compute visual complexity	+compute
	-visually complicate	visually complicate	-visually complicate	-visually complicate	- operationally complicate	visually complicate	reduce reliability		visually complicate
	+qualify	+expose	capture	+assure	+assure	+capture	+predict	+capture	

Table F.6 captures the AIC complexity field for the AVOIDDS problem domain, where all interactions are accounted for. Note, we also included the architect as part of the problem.

Appendix F

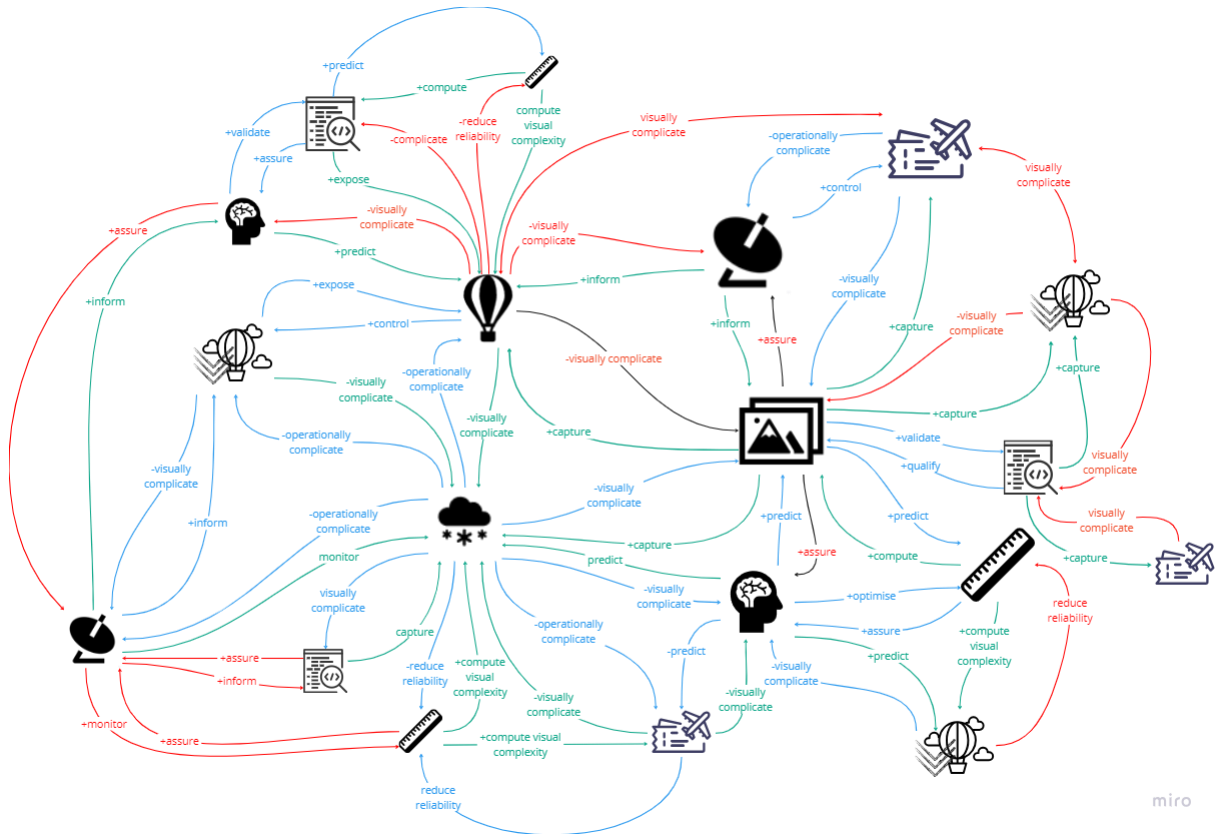


Figure F.1 AIC complexity field of AVOIDDS problem

The icons are defined in the following table, which captures the main sources of complexity:

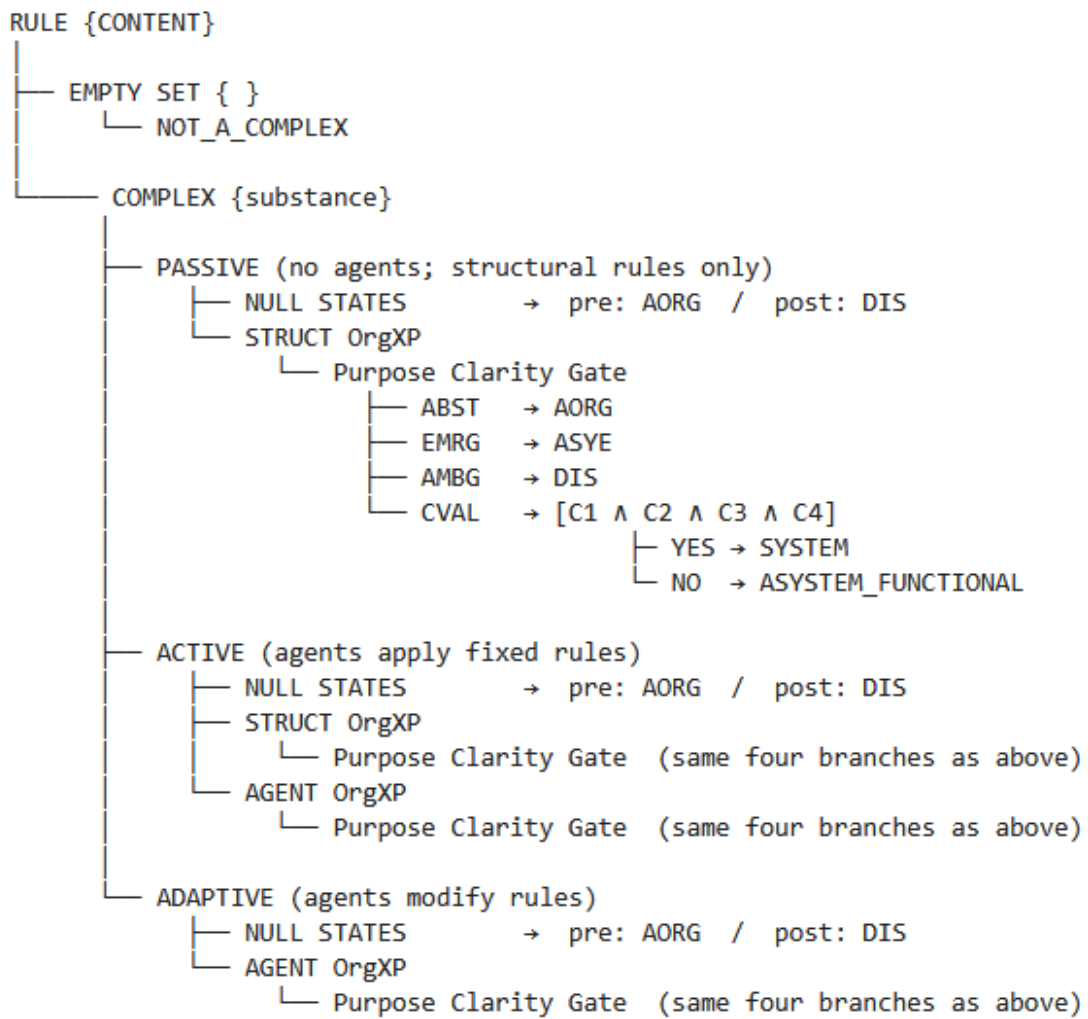
Table F.7 AVOIDDS problem complexes icons definition

Scenario	Icon
other_flying_objects	
weather_conditions_variety_definition	
accurate_architect_predictive_mental_model_ArcPM	
informed_air_traffic_controller	
avp_development_datasets (avp stands for aircraft avoidance perception system)	
reliable_correct_target_object_recognition_confidence_scores	
critical_data_accuracy	

visual_traffic_density_and_patterns	
other_flying_objects_visual_complexity_parameters	

Overall, the AIC-Matrix framework provides a structured approach to resolving complicated interactions in a long-tailed complexity field (abnormal complexity). It enables the architect to predict how desired systems should behave, predict Black Swan Scenarios and enables more resilient assumptions in dynamic and uncertain environments.

F.2 General Organisations Assumptions



Appendix F

#	Concept	Layer	Definition
0	RULE	Root	The universal rule set
1	CONTENT	Root	The universal set of all possible entities. Contains both the empty set and all non-empty collections of substance. The sole branching criterion is whether substance is present.
2	EMPTY SET (NOT_A_COMPLEX)	Root	A content set with zero substance , no physical entities, knowledge entities, or rules. By definition, it is not a complex and is excluded from all further classification.
3	COMPLEX	Type	A non-empty collection of substance (at least one physical entity, knowledge entity, or rule). The minimal unit of analysis in the ontology.
4	PASSIVE	ComplexType	A complex whose substance contains no agents , entities capable of applying or modifying rules. Compartment is governed exclusively by structural (mechanistic) rules.
5	ACTIVE	ComplexType	A complex containing agents capable of applying a fixed rule set. The agents do not modify the rules; their capability is rule-application only.
6	ADAPTIVE	ComplexType	A complex containing agents capable of modifying the operative rule set itself. Encompasses all capability present in ACTIVE, extended by rule-modification capacity.
7	NULL STATES	OrgXP	The two states in which no operative rule set currently produces observable behavioural patterns. Subdivided into NULL_PRE_ORG and NULL_POST_ORG by the historical trajectory of the complex.
8	NULL_PRE_ORG (pre→AORG)	OrgXP	A null state in which no OrgXP has ever existed for this complex. There is no historical rule framework. Outcome is necessarily AORGANISATIONAL.
9	NULL_POST_ORG (post→DIS)	OrgXP	A null state in which a formerly operative OrgXP has collapsed and is no longer producing observable patterns. The prior existence of a rule framework is the distinguishing fact. Outcome is DISORG.
10	STRUCTURAL OrgXP (STRUCT)	OrgXP	An operative rule set in which behavioural patterns are produced by physical, logical, or mechanistic necessity , without requiring a perception–response loop. Available to PASSIVE and ACTIVE complexes. Incompatible with ADAPTIVE.
11	AGENTIVE OrgXP (AGENT)	OrgXP	An operative rule set in which behavioural patterns are produced through agents executing a perception–response cycle. Requires agents; available to ACTIVE and ADAPTIVE complexes only.
12	PURPOSE CLARITY	Purpose	The property of the attributed purpose of a complex, ranging from complete absence through emergent direction to internal contradiction to coherent validation. Acts as the primary gate before the 4-test.
13	ABSENT (ABST)	PurposeClarity	A state in which no purpose has been attributed or is attributable to the complex. The complex produces observable patterns (if OrgXP exists) but with no directional goal. Outcome: AORGANISATIONAL.

Appendix F

1 4	EMERGING (EMRG)	PurposeClarity	A state in which a directional purpose is nascent , partially identifiable but not yet operationalised or validated. The goal exists as a trajectory, not a confirmed achievement. Outcome: ASYSTEM_EMBRYONIC.
1 5	AMBIGUOUS (AMBG)	PurposeClarity	A state in which the attributed purpose is internally contradictory: it simultaneously requires mutually exclusive conditions ($P_a \wedge \neg P_b$). No globally coherent C/D/N rule partition exists (AX-RT-8). Outcome: DISORG.
1 6	CLEAR_VALIDATED (CVAL)	PurposeClarity	A state in which the attributed purpose is internally coherent and empirically confirmed , the complex demonstrably achieves its goal. Necessary but not sufficient condition for SYSTEM. Gate to the 4-test.
1 7	AORGANISATIONAL (AORG)	Outcome	A complex in which no organisational experience exists or has ever existed. Arises from: (a) NULL_PRE_ORG, or (b) any operative OrgXP with ABSENT purpose. Not pathological , it is a structural pre-organisational state.
1 8	DISORG	Outcome	A complex in an incoherent operational state arising from one of two structurally distinct paths: (A) NULL_POST_ORG , operational collapse of a formerly operative rule framework; (B) AMBIGUOUS purpose , active rule set operating toward a self-contradictory goal. Recovery path differs by subtype (AX-RT-10).
1 9	ASYSTEM_EMBRYONIC (ASYE)	Outcome	A complex with a valid operative OrgXP and an EMERGING purpose that has not yet been operationalised or validated. A positive aspirational state; development trajectory is toward ASYSTEM_FUNCTIONAL or SYSTEM.
2 0	4-TEST GATE	Gate	A four-criterion logical conjunction applied exclusively to complexes at CLEAR_VALIDATED purpose. The test evaluates: C1 (entity definability), C2 (OrgXP validity), C3 (cooperative rule dominance), C4 (perturbation convergence). $SYSTEM \Leftrightarrow C1 \wedge C2 \wedge C3 \wedge C4$.
2 1	C1 (Entity Definability)	4-Test	All operationally relevant entities , those within the Architect-specified system boundary , are formally definable. Prerequisite for C2 and C3 (AX-4T-5).
2 2	C2 (OrgXP Validity)	4-Test	The operative rule set is: (a) predictable in outcomes, (b) purposively aligned with the attributed goal, (c) formally definable, and (d) independently validateable. Strictly stronger than CLEAR_VALIDATED purpose alone.
2 3	C3 (Cooperative Dominance)	4-Test	The aggregate impact of purpose-cooperative rules strictly exceeds the combined impact of purpose-destructive and purpose-neutral rules: $impact(cooperative) > impact(destructive) + impact(neutral)$. The impact measure is domain-specific (Architect-specified).
2 4	C4 (Perturbation Convergence)	4-Test	When perturbed within the Architect-specified operational envelope, the complex converges to a stable, predictable order. For ADAPTIVE complexes: extended to meta-stability , a stable attractor in the

			rule-modification space (AX-4T-4). Logically independent of C1, C2, C3.
2 5	ASYSTEM_FUNCTIONAL	Outcome	A complex with a valid operative OrgXP and CLEAR_VALIDATED purpose that fails at least one of C1, C2, C3, or C4. Operationally present but below SYSTEM standard. Distinct from DISORG: purpose is coherent; only the implementation falls short.
2 6	SYSTEM	Outcome	A non-empty complex with a formally specified boundary, a CLEAR_VALIDATED purpose, and an operative OrgXP that simultaneously satisfies all four criteria $C1 \wedge C2 \wedge C3 \wedge C4$ within a defined assessment epoch. The unique terminal outcome that denotes full organisational realisation.

Complex is a non-empty substance, comprising physical entities, knowledge entities, or both, together with any rules that govern the interactions of those entities, whether those rules are fully specifiable or not. Physical entities are tangible, spatially locatable things (infrastructure, agents, natural phenomena); knowledge entities are abstract, informationally defined things (rules, models, beliefs, specifications). By strict negation, the only thing that is not a complex is the empty set: a set containing neither entities nor rules. A complex has no minimum size or formal specificity requirement; a train-track zone, a single organism, and a mathematical proof are all complexes.

Substance is the non-empty content of a complex that gives it definable existence. Substance can be gained, lost, or maintained depending on the operative rules of the complex.

Complexes are classified by **ComplexType**, which is a capability classification, it describes what a complex's agents are structurally capable of doing, not what they are currently doing. Three types are defined:

- A **Passive complex** is one that contains no agentive entities; any organisational patterns arise exclusively from physical or logical necessity. In AIC schema, a passive complex is captured by a rectangular shape.
- An **Active complex** is one that contains at least one agentive entity structured to apply a fixed rule set. In AIC schema, an Active complex is captured by an octangular shape.
- An **Adaptive complex** is one that contains at least one agentive entity structured to modify its own rule application, adding, removing, or altering rules through experience or feedback. In AIC schema, an Active complex is captured by an oval shape.

Organisational Experience (OrgXP): The Organisational Experience of a complex is the operative rule set of that complex, the set of rules that are currently active and producing observable behavioural patterns within it. OrgXP is classified by its execution mode:

- **A Structural OrgXP** is an operative rule set that holds by physical or logical necessity, requiring no agentic application. It applies to passive and active complexes (e.g. thermodynamic equilibration, crystal growth geometry, a thermostat's bimetallic switching circuit).
- **An Agentic OrgXP** is an operative rule set that is executed by agents through a perception–response loop, in response to perturbations. It applies to active and adaptive complexes (e.g. a compliance team enforcing regulatory rules, an immune cell responding to a pathogen).

A passive complex cannot have an agentic OrgXP (no agents are present to apply rules); an adaptive complex cannot have a structural OrgXP (rules that are modified by agents are, by definition, not operating by physical necessity).

Two null states are defined independently of execution mode:

- **NULL_PRE_ORG** is the state of a complex in which an operative rule set has never been established. This is a pre-organisational state; no history of governance exists.
- **NULL_POST_ORG** is the state of a complex in which a previously operative rule set has since become absent, obsolete, or internally inconsistent. This is the state of organisational collapse; an organisational history exists but has ended.

Rules within an operative OrgXP are classified along two orthogonal axes relative to the attributed purpose: (i) the purpose-relative axis, each rule is cooperative (contributes positively to the attributed purpose), destructive (works against it), or neutral (contributes nothing net); and (ii) the mutual-interaction axis, each rule is enabling, inhibiting, or independent with respect to the execution of other rules. These axes are logically independent: a rule that is cooperative toward the purpose may simultaneously inhibit the execution of another cooperative rule.

Purpose Clarity: The attributed purpose of a complex is the goal toward which the OrgXP is directed. Purpose clarity is classified on a single axis of increasing specificity and coherence:

- **Absent**, no purpose is attributable to the complex whatsoever.
- **Emerging**, a purpose direction has been stated but has not yet been operationalised into rules or mechanisms for achieving it.
- **Ambiguous**, a purpose has been operationalised, but the purpose statement is internally contradictory, containing self-defeating requirements that cannot all be satisfied simultaneously.

- **Clear-Validated**, a purpose that is coherent, fully operationalised, and empirically confirmed as achievable by the complex under operational conditions.

System: A system is a complex with a clear-validated attributed purpose that simultaneously satisfies all four of the following criteria:

- **(C1) Entity Definability:** all operationally relevant entities within the independently specified system boundary are formally definable. An entity is operationally relevant if it participates in at least one operative rule of the OrgXP; the system boundary must be specified independently of the rule set to avoid circularity.
- **(C2) OrgXP Validity:** the OrgXP is predictable, functionally aligned with the attributed purpose, and its rules are both formally definable and independently validateable through a declared verification procedure. This is a mechanism-level criterion; it is strictly stronger than clear-validated purpose alone, which is an outcome-level criterion.
- **(C3) Cooperative Dominance:** the aggregate impact of the cooperative rules strictly exceeds the combined aggregate impact of the destructive and neutral rules. The impact measure is domain-specific and must be independently specified.
- **(C4) Perturbation Convergence:** when perturbed within an independently and non-circularly specified operational envelope, the complex converges to a stable, predictable order (static or dynamic). For adaptive complexes with continuous rule modification, convergence is satisfied by meta-stability: a stable, independently specifiable attractor in the rule-modification space (second-order stability).

All four criteria must hold simultaneously. Failure of any one criterion produces an asystem, not a system. The application of criteria C1–C4 (the 4-test) is gated on clear-validated purpose: the test is undefined under absent, emerging, or ambiguous purpose, because C2 requires a coherent attributable purpose and C3 requires a globally consistent purpose relative to which rules can be classified as cooperative, destructive, or neutral.

Asystem: An asystem is a complex that is not a system but possesses a coherent purpose direction. Two structurally distinct subtypes are defined:

- **An ASYSTEM_EMBRYONIC** is a complex with a valid operative OrgXP and an emerging purpose: a purpose direction exists and is stated, but has not yet been operationalised into rules or a functional mechanism for achieving it. The complex is aspirational but not yet operationally organised.

- **An ASYSTEM_FUNCTIONAL** is a complex with a valid operative OrgXP and a clear-validated purpose that fails at least one of the four system criteria C1–C4. The operational framework exists and the purpose is coherent, but the implementation falls short of the system standard. The specific criterion that fails determines the appropriate corrective intervention; only failure of C3 is directly addressable through adjustment of the rule-type balance.

Disorganisation Experience (DISORG): A Disorganisation Experience is the structural state of a complex in which organisational coherence is absent through one of two distinct mechanisms:

- **DISORG(NULL_POST_ORG):** the complex's previously operative rule framework has collapsed, no currently operative rules exist. The non-system state reflects operational collapse. Recovery requires structural re-establishment of operative rules.
- **DISORG(AMBIGUOUS):** the complex has an operative rule set directed toward a purpose that is internally contradictory. The OrgXP is present and rules are executing, but the purpose-level incoherence makes coherent goal-directed functioning impossible, regardless of how the rules are composed or how cooperative they are individually. Recovery requires resolving the purpose contradiction; adjusting the rule-type balance is insufficient (since C3 cannot even be evaluated under ambiguous purpose).

DISORG is a structural property of the complex's OrgXP or purpose state, not a property of the complex's rule-type composition. A complex may be in DISORG with any combination of cooperative, destructive, or neutral rules. Misidentifying the active DISORG subtype leads to the wrong recovery strategy: applying purpose-level intervention to a collapsed OrgXP achieves nothing; applying rule-level restructuring to a contradictory purpose intensifies purposively incoherent activity without resolving the structural fault.

Predictability: Predictability is a property of the OrgXP: it is the degree to which the temporal evolution of the complex's behavioural patterns can be determined in advance from knowledge of its current rule set, initial state, and perturbation. A maximally predictable OrgXP is one for which the outcome of any perturbation within the operational envelope can be determined prior to observation. Predictability is one of the four sub-conditions required for OrgXP Validity (C2) and is therefore a necessary but not sufficient condition for system status.

F.2.1 ComplexType Enum

What it classifies: The structural capability of the agents within a complex. This is a *capability*-typed classification, not an operational-state classification (see AX-C11-3). A complex's

ComplexType tells us what its agents are *structured to do*, regardless of whether they are currently doing it.

a) EMPTY_SET

A trivial case: the content under consideration contains no substance at all. It does not constitute a complex. **Outcome:** NOT_A_COMPLEX always.

Example: A null set, an unfilled container, a defined but uninstantiated class with no members.

How derived: Logically prior to all other classification, any classification procedure must gate on "does non-empty substance exist?" before proceeding.

b) PASSIVE

A complex whose substance is entirely non-agentive. There are no entities with the capacity to perceive, decide, or apply rules. Any organisational patterns that exist arise from physical or logical necessity, gravity, thermodynamics, chemical bonding, mathematical structure.

Example: A crystal lattice, a river drainage network, a protein folding into its minimum-energy conformation, a pile of sand forming an angle of repose.

How derived: From AX-C12-1: "PASSIVE means NO agentive entities are present." The absence of agents eliminates the possibility of AGENTIVE OrgXP (AX-C13-3) and also eliminates ADAPTIVE ComplexType (which requires rule-modification capability, itself agentive).

Key implication: A PASSIVE complex can still achieve SYSTEM status, its structural rules may satisfy all four 4-test criteria under a teleonomic (purpose-attributed) rather than teleological (purpose-intended) interpretation.

c) ACTIVE

A complex containing at least one agentive entity that applies a **fixed** rule set. The agents have a repertoire of operational responses and execute them, but do not modify the rule set itself.

Example: A thermostat applying a fixed temperature-response rule ("if $T < \text{setpoint}$, activate heater"), a classical bureaucracy following a fixed procedure manual, a computer executing a non-self-modifying program.

How derived: From AX-C12-2: "ACTIVE means ≥ 1 agentive entity is present AND the entity applies a fixed rule set." Critically, ACTIVE complexes are compatible with *both* STRUCTURAL and

AGENTIVE OrgXP (AX-C13-5) , the Architect must determine which applies based on whether rule execution is mechanistically necessitated or perception-response driven.

A common confusion avoided: ACTIVE + NULL_PRE_ORG is *not* paradoxical. Under capability-typing (AX-C11-3), ACTIVE says the agents *could* apply rules , but they may never have been given any rules to apply. Outcome: AORGANISATIONAL.

d) ADAPTIVE

A complex containing at least one agentive entity that can *modify* rule application , adding, removing, or altering rules based on experience, feedback, or learning.

Example: A biological immune system (generates new antibody rules in response to pathogens), an organisation with a continuous improvement culture, a machine learning model updating its weights, an adaptive governance system revising its own procedures.

How derived: From AX-C12-3: "ADAPTIVE means ≥ 1 agentive entity is present AND the entity can MODIFY rule application." Rule modification implies agents (you cannot modify a rule without agency), so ADAPTIVE + STRUCTURAL OrgXP is logically impossible (AX-C13-4): structural rules operate by necessity, not by agentive modification.

Key complication: ADAPTIVE complexes may never settle into a fixed rule set, requiring C4 of the 4-test to be interpreted as *meta-stability* (second-order attractor in rule-modification space) rather than first-order convergence (AX-4T-4, H6).

F.2.2 OrgXPType Enum

What it classifies: The *organisational experience* (OrgXP) , the mode by which operative rules are established and executed within a complex.

a) NULL_PRE_ORG

The complex has **never** been governed by operative rules. No organisational experience has ever existed.

Example: A newly formed group of people with no shared norms, procedures, or practices yet. A newly created database with no records or constraints. A chemical solution before any reaction has begun.

How derived: From AX-C11-1: "NULL_PRE_ORG: complex was NEVER governed by rules → outcome is AORGANISATIONAL." The *pre* prefix signals temporal precedence, there has been no organisational history whatsoever.

Outcome always: AORGANISATIONAL

b) NULL_POST_ORG

The complex **previously** had operative rules, but those rules have become absent, obsolete, or internally inconsistent. The organisational experience has *collapsed*.

Example: A company whose procedures have fallen into disuse after a major restructuring. A biological organism in a pathological state where homeostatic regulation has broken down. A governance system whose laws have become contradictory and unenforced.

How derived: From AX-C11-2: "NULL_POST_ORG: complex previously had operative rules that are now absent/obsolete/inconsistent → outcome is DISORG." The *post* prefix signals that an organisational history exists but has ended. This is structurally distinguishable from NULL_PRE_ORG because recovery requires *re-establishing* rules, not *establishing them for the first time*.

Outcome always: DISORG

c) STRUCTURAL

Operative rules arise from **physical or logical necessity**, they require no agentic application. The OrgXP is constituted by the structural constraints of the medium itself.

Example: The laws of thermodynamics governing energy flow in an ecosystem. The geometric constraints governing crystal growth. The axioms of arithmetic governing number theory. Traffic flow patterns emerging from road geometry.

How derived: From AX-C13-1: "STRUCTURAL OrgXP: rules operate by physical/logical necessity; no agentic application required. Applies to PASSIVE and ACTIVE complexes." For ACTIVE complexes, the distinguishing criterion (AX-C13-5) is whether the rule is *mechanistically necessitated* (STRUCTURAL) or *perception-response driven* (AGENTIVE).

d) AGENTIVE

Operative rules are applied by agents **in response to perceived perturbations**, the OrgXP exists because agents are actively executing it.

Example: A police force enforcing laws. A company's compliance team following regulatory rules. A doctor applying diagnostic protocols in response to patient symptoms. A robot executing a decision-policy.

How derived: From AX-C13-2: "AGENTIVE OrgXP: rules are applied by agents in response to perceived perturbations. Applies to ACTIVE and ADAPTIVE complexes." PASSIVE complexes cannot have AGENTIVE OrgXP (AX-C13-3) because there are no agents to apply rules.

F.2.3 PurposeClarity Enum

What it classifies: The degree to which the complex's attributed purpose is specified, coherent, and empirically validated. This is the *second* major axis of classification, operating independently of ComplexType.

a) ABSENT

No purpose is attributable to the complex at all , not even a directional aspiration.

Example: Raw undifferentiated matter. A random number generator. Background cosmic noise. An undirected assembly of components with no intended function.

How derived: From AX-C14-1: "ABSENT purpose: no purpose attributable → AORGANISATIONAL (not DISORG)." The distinction from DISORG is critical: DISORG implies a complex that had/has purpose but it is incoherent. ABSENT means purpose is simply not a concept that applies , there is nothing to be incoherent about.

Outcome: AORGANISATIONAL

b) EMERGING

A purpose **direction** exists and is stated, but has not yet been operationalised into rules, procedures, or a mechanism for achieving it.

Example: A startup with a clear mission statement ("make space travel accessible") but no operational infrastructure yet. A social movement with a defined aspiration but no organisational structure. An early-stage research programme with a directional hypothesis but no operational protocol.

How derived: From AX-C14-2: "EMERGING purpose: purpose stated, not yet operationalised → ASYSTEM (has direction; not DISORG)." The complex is organisationally aspirational but not yet

functionally organised. It is neither incoherent (DISORG) nor fully operational (ASYSTEM_FUNCTIONAL or SYSTEM).

Outcome: ASYSTEM_EMBRYONIC

c) **AMBIGUOUS**

The purpose has been operationalised (rules exist attempting to achieve it) but the purpose itself is internally **contradictory**, it contains self-defeating requirements.

Example: An organisation simultaneously required to "maximise cost savings" and "provide unlimited service quality" with no principled trade-off mechanism. A law that requires both "protect privacy" and "provide complete transparency" in all cases. A robot instructed to "always minimise energy consumption" and "always respond immediately to all queries."

How derived: From AX-C14-3: "AMBIGUOUS purpose: operationalised but internally contradictory → DISORG." This is one of the most important findings: the complex may have *working rules* and an *operative OrgXP*, yet it is in DISORG because the *purpose itself* is incoherent. No rule-balance adjustment can resolve this (H18); only purpose clarification can recover it (H19).

Outcome: DISORG (regardless of rule balance, see H16)

d) **CLEAR_VALIDATED**

The purpose is coherent, operationalised, and has been **empirically tested**, the complex demonstrably achieves what it is organised to achieve.

Example: A working commercial aircraft (purpose: safe passenger transport at scale, validated through certification and operational history). A proven vaccine (purpose: induce immunity, validated through clinical trials). A working operating system (purpose: manage hardware resources for applications, validated through use).

How derived: From AX-C14-4: "CLEAR_VALIDATED purpose: operationalised AND empirically tested → eligible for SYSTEM/ASYSTEM determination via the 4-test." This is the *gateway* to SYSTEM status, no complex can be classified SYSTEM without first achieving CLEAR_VALIDATED purpose.

Outcome: Determined by the 4-test ($C1 \wedge C2 \wedge C3 \wedge C4 \rightarrow \text{SYSTEM}$; any failure → ASYSTEM_FUNCTIONAL)

F.2.4 Outcome Enum

The outcome enum captures the full classification space. These are the possible conclusions of applying the complete ontology to a complex.

Outcome	Meaning
NOT_A_COMPLEX	Content is the empty set , no complex to classify
AORGANISATIONAL	Complex exists but has no organisational experience (NULL_PRE_ORG or ABSENT purpose)
DISORG	Complex has incoherent/collapsed organisational structure (NULL_POST_ORG, or AMBIGUOUS purpose)
ASYSTEM_EMBRYONIC	Complex has valid OrgXP + EMERGING purpose , aspirational, not yet operational
ASYSTEM_FUNCTIONAL	Complex has valid OrgXP + CLEAR_VALIDATED purpose + fails ≥ 1 of C1-C4
ASYSTEM	Umbrella term for both EMBRYONIC and FUNCTIONAL (avoid in analytical precision contexts)
SYSTEM	Complex has valid OrgXP + CLEAR_VALIDATED purpose + all four 4-test criteria satisfied
SYSTEM_OR_ASYS	Intermediate state when the 4-test is not yet defined (resolved in Pass 3)
UNDEFINED	Edge case

Key finding (H15): Only ALL_COOPERATIVE and COOPERATIVE_DOM can produce SYSTEM , they are the only configurations where C3 (cooperative rule dominance) *may* hold. All other configurations at CLEAR_VALIDATED purpose produce ASYSTEM_FUNCTIONAL at best.

Key finding (H13): Even ALL_COOPERATIVE can produce DISORG , if the purpose is AMBIGUOUS, cooperative rules are cooperating toward a contradictory goal. Rule cooperation is locally coherent but globally purposively incoherent.

F.3 Stage 1: Problematic Operational Environment Unforeseeability Reduction STCoTs

F.3.1 STCoT_1 Unforeseeability Systemic predictive reasoner

Rule A: Highly Complicated Heterogeneous Complexity:

When an operational environment simultaneously contains all of the following complex types:

Complex Type	Examples
Living agents	Humans, animals, biological systems
Natural systems	Weather, terrain, electromagnetic environment
Computerised agents	AI systems, autonomous agents, software
Deterministic non-agentic systems	Mechanical structures, fixed infrastructure

Then any perturbation in one complex propagates non-linearly across all others. The result is a Compounded Epistemic Uncertainty Problem (CUPE): emergent atypical scenarios whose causal pathways exceed the analyst's predictive mental model, at a rate that increases with the number of heterogeneous types present.

The mathematical intuition: if n is the number of distinct complex types and k is the number of individual complexes, the emergent scenario (possible interaction rules) space grows at least as $O(k^n)$, far beyond any exhaustive enumeration.

Rule B: Atypical Causality Inversion Principle:

This is the most subtle and powerful rule. It states:

If a typical obstructive interaction prevents complex C_2 from manifesting its primary purpose, AND C_2 possesses at least one supportive interaction directed *toward* the very complex C_1 that is obstructing it, THEN an atypical self-defeating paradox exists: C_2 's own actions inadvertently *support* C_1 's obstruction of C_2 .

In plain language: **the victim helps the threat**. This is structurally invisible under normal attentional distributions because the analyst's mental model tracks forward causality ("C1 threatens C2") but not the paradoxical feedback loop ("C2 enables its own harm").

Illustrative analogy: A pilot (C_2) whose collision avoidance system (C_1) is generating false alerts. The pilot, in attempting to manage cockpit workload (pursuing their own primary purpose), mutes the TCAS alerts, thereby *supporting* the very condition (unawareness of intruders) that the TCAS was preventing. The pilot's coping mechanism becomes the threat pathway.

e) The STCoT_1 Process: How It Works

STCoT_1 is a structured two-phase predictive thinking process (ArcPT):

INPUT: Problem Brief (operational scenario description)

GSM ANCHOR: Rules A (Heterogeneity) + B (Causality Inversion)

PTP 1 , FORESEEABLE COMPLEXITY RESOLUTION

Step 1.1: Define the scenario + undesirable events

Step 1.2: Generate descriptive image of the scenario

Step 1.3: Enumerate all complexes (static heterogeneity)

Step 1.4: Define Supra-Complexes and their PrimePs

PTP 2 , SOURCE-SINK COMPLICATEDNESS RESOLUTION

Step 2.1: TYPICAL scenarios , Action Matrix

(a) Source loses ability to manifest purpose

(b) Sink loses ability to manifest purpose

(c) Source + Sink obstruct each other

Step 2.2: UNFORESEEABLE scenarios , Causality Inversion

"Sink deceptively allows/supports Source to..."

OUTPUT: Architect Assertion , set of unforeseeable scenarios
grounded in Rules A + B

f) Practitioner Step-by-step implementation

STEP 0 Write a Problem Brief (one paragraph: what is the scenario?)

STEP 1 List all complexes , assign each:

- An ID, a Kind (Living/Natural/Computerised/Deterministic)
- A Primary Purpose (PrimeP)

STEP 2 Identify Supra-Complexes and their higher-order PrimePs

STEP 3 Build the Action Matrix , for every source-sink pair, define:

- The interaction direction and polarity (+/-/0)
- A plain-language description of the action

STEP 4 For each pair, generate 3 TYPICAL scenarios:

- (a) Source fails to manifest purpose
- (b) Sink fails to manifest purpose
- (c) Source and sink mutually obstruct each other

STEP 5 For each interaction, generate 1 UNFORESEEABLE scenario: "Sink DECEPTIVELY ALLOWS/SUPPORTS Source to [harm Sink's purpose]" → This is the mandatory inversion , the core of STCoT_1 STEP 6 Check Architect Assertion 1.4: - All complexes have PrimePs? - All pairs covered in the matrix? - All three typical patterns present per pair? - One inversion scenario per interaction? STEP 7 Document output as candidate unforeseeable scenarios grounded in Rules A and B
--

The critical innovation is Step 2.2. By systematically inverting every source-sink interaction, turning obstructive interactions into deceptive support, the analyst generates scenarios that are structurally invisible under normal forward-causal reasoning but formally derivable from Rule B. The completion criterion is explicit: the step is complete when, for every paired interaction, an atypical confusing interaction has been defined in which the sink deceptively allows or supports the source to act in a way that defeats the sink's own purpose.

g) What STCoT_1 Produces: Illustrated Output

For the mid-air collision case study, the key output from Step 2.2, the analytically novel results are:

#	Causality Inversion Scenario	Why Invisible Under Normal ATD
1	Pilot deceptively supports TCAS to be muted: workload-driven silencing of collision alerts enables an intruder approach to be undetected	Analyst tracks TCAS→Pilot (forward); misses Pilot→TCAS feedback
2	Pilot deceptively supports ATC to issue risky clearances: blind compliance defers all authority to ATC, even when ATC radar is degraded	Analyst treats ATC→Pilot as protective; misses Pilot's authority surrender
3	Ownship deceptively supports Weather to cause hazard: seeking turbulent airspace to avoid traffic inadvertently degrades collision avoidance, where it is most needed	Analyst models weather as a passive threat; misses pilot-weather intentional coupling
4	TCAS deceptively supports Intruder approach: mis-synchronised RAs direct ownship toward intruder flight path	Analyst treats TCAS→pilot as protective; misses TCAS→intruder enabling pathway

h) Full STCoT_1

Table F.8 STCoT_1 for Predicting Unsafe Behaviours

STCoT 1	Unforeseeability Systemic predictive reasoner
Input	Problem brief

General Systems Model	General rule A: Highly Complicated Heterogeneous Complexity: IF an operational environment contains all of the following types of complexes (static heterogeneity) {living agents, natural systems, computerised agents, deterministic non-agentic systems} within an open bounded environment, THEN any OrgXP perturbation in one complex propagates non-linearly across the scenario, producing emergent atypical Scenarios whose causal pathways exceed the inquirer's predictive mental mode to explain, at a rate that increases with the number of Heterogeneous complex types present. LEADING TO unforeseeable Compounded Epistemic Uncertainty Problem General rule B: Atypical Causality Inversion Principle: IF a typical obstructive interaction $[[C1]_{-action}]_{C2}$ prevents sink complex C2 from manifesting its PrimeP, $[[C1]_{-action}]_{C2} \rightarrow \text{not}\{C2 \text{ PrimeP}\}$ AND C2 possesses at least one appreciative. control or influence interaction directed toward C1, THEN an atypical causality inversion exists in which C2's own interactions redirect to support C1's obstruction of C2's PrimeP, producing a self-defeating paradoxical OrgXP in C2 that is non-salient to the inquirer under a normal Architect Attention Distribution (ATD); $[[C2]_{+action}]_{C1} \rightarrow \text{not}\{C2 \text{ PrimeP}\}$. LEADING TO unforeseeable Compounded Epistemic Uncertainty Problem
Predictive Thinking Approach (ArcPT)	Predictive query 1.1: Which undesirable behaviours, actions, or interactions can plausibly (i) degrade the predictability of the scenario behaviour, thus degrading the architect's mental model viability, (ii) produce source-to-sink coupling causing purpose loss, and (iii) generate high-impact outcomes unexplainable through expected dynamics? Guiding Prompt 1.2: Framing principle: The operational domain is treated as open, and atypical causality is assumed to be possible. Indirectly coupled events are retained for analysis. Atypical causality is the “unknown” and can be validated after a definition of typical causality. PTP 1: Foreseeable Complexity Resolution Step 1.1) Define the input high-level scenario, undesirable or problematic Scenarios [assumption: what is known in this definition is typical], Step 1.2) Generate a descriptive image that visualises the undesirable behaviour [assumption: what is visible in this depiction is knowable]. Step 1.3) Define the set of complexes of the scenario's typical complexity. (initial static heterogeneity). Step 1.4) Define the Supra Complexes and their PrimePs. (initial static heterogeneity). PTP 2: General Source-Sink Rules Complicatedness Resolution (possible rules that underpin complexity) Step 2.1) Predict typical undesirable scenarios: Apply the Action matrix to pair each complex with another. Predict undesirable scenarios based on any of the following general scenario models: For each pair (source, therefore sink): [assumption: what is defined in this step is a model of typical and typical]. This step analyses the rules that underpin typical undesirable complexity. - Source loses the ability to manifest purpose. - Sink loses the ability to manifest purpose. - Sink and Source obstruct each other's abilities to manifest respective purposes.

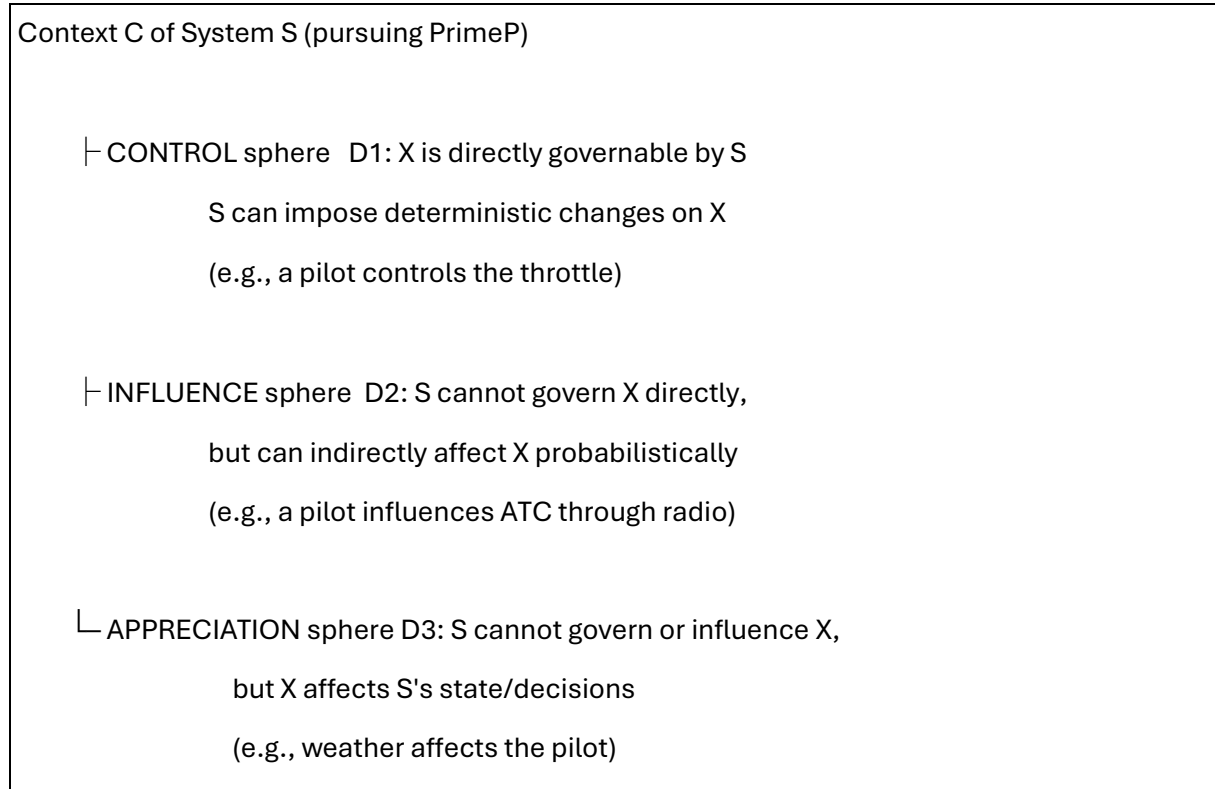
	Step 2.2) Predict unforeseeable scenarios: for each interaction, predict a possible confusing scenario that would be deemed as atypical, or abnormal and does not align with expected purposes. Turn undesirable interactions into supportive ones, so that the sink is designed or intended to support the source's actions. Furthermore, start the thought step with: Sink deceptively allow/support source to ... [assumption: what is defined in this step is atypical and unforeseeable without it]. This step analyses the rules that underpin unforeseeable complexity. Step completion criteria 1.3: The step is regarded as complete when a problem Scenario is described, including undesirable impact. • An input scenario is defined. • A list of notable complexes. • A notable purpose associated with each complex. • An action matrix that maps all complexes to all. For each paired interaction, an atypical confusing interaction is defined.
Output Prediction	Architect assertion 1.4: The Architect asserts that the output predictions are pragmatically consistent, structurally complete, and are acceptably likely and impactful, with respect to the validity and applicability of general rules A and B. Unforeseeable (confusing) scenarios: • Ownship deceptively allows/supports Intruder to approach unnoticed by muting TCAS/ADS-B, enabling unchallenged crossing and potential strike. • Ownship deceptively allows/supports ATC to issue risky clearances by blindly following instructions. Ownship deceptively allows/support Environment to cause hazards by intentionally seeking environmental turbulence, since less traffic is expected in such environments, due to assuming less risk of mid-air collision.

F.3.2 STCoT_2 AIC-Complexity Systemic predictive reasoner (Forward and Backwards-Feed)

STCoT_2 is anchored in five formal IF-THEN rules drawn from the General Systems Model (GSM). These are the logical guarantors of the predictions it generates, any prediction STCoT_2 produces is justified by appeal to at least one of these rules:

Rule	Name	Core Claim
A	Ordered Relational System Definition (Smith)	A complex with a defined purpose and ordered relationships constitutes a system
B	Contextual Embeddedness of Purpose (Smith)	Purpose is always pursued within a context containing external phenomena
C	Appreciative Awareness as Behavioural Prerequisite (Smith)	Purposeful behaviour requires appreciative awareness of both purpose and context
D	AIC Relational Decomposition of Context (Smith)	Any context can be exhaustively decomposed into Control, Influence, Appreciation
E	Recursive System Hierarchy (Mobus)	Systems contain subsystems and belong to supra-systems; relational inheritance applies

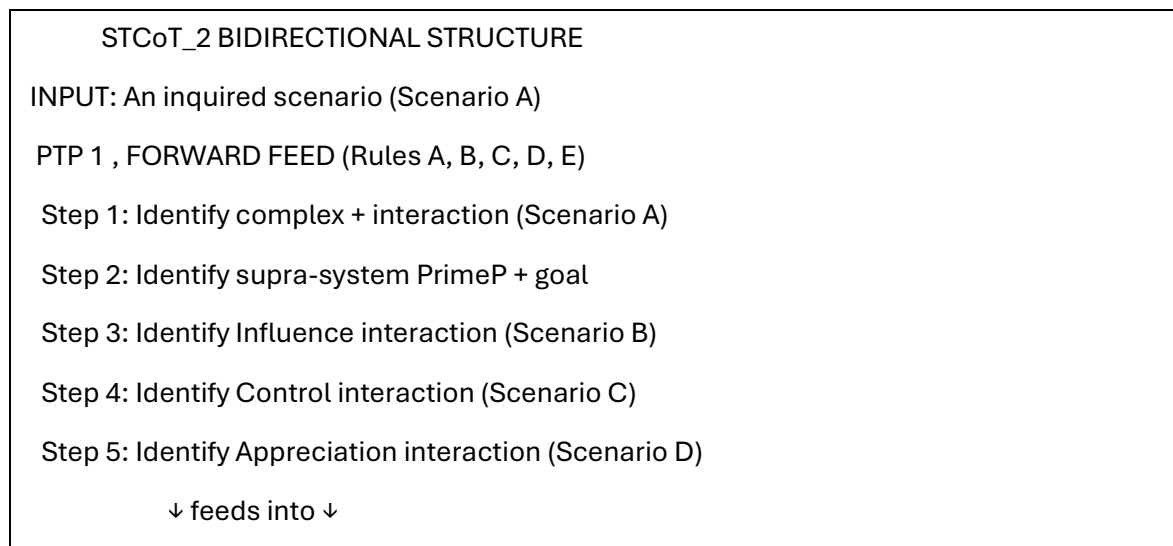
The exhaustiveness condition of Rule D is the most operationally important: every element of the context must be classifiable as Control, Influence, or Appreciation. If it cannot be classified, the model is structurally incomplete and harbours unmitigated epistemic uncertainty. The central analytical tool of STCoT_2 is the AIC decomposition of a system's context with respect to its primary purpose:



The key insight: these three categories are exhaustive and mutually exclusive. Every interaction between a system and its context must fit exactly one category. This gives STCoT_2 its structural completeness guarantee.

a) The STCoT_2 Structure: Nine-Step Bidirectional Process

STCoT_2 consists of two Predictive Thinking Pipelines (PTPs):



PTP 2 , BACKWARD FEED (Rule E: relational inheritance)

Step 6: Identify Influencing Scenarios (Scenario X)

Step 7: Identify Controlling Scenarios (Scenario Y)

Step 8: Identify Appreciative Scenarios (Scenario Z)

Step 9: Define A's AIC feedback to X, Y, Z

OUTPUT: Architect Assertion , complete AIC interaction map

Completion criterion: Each step is complete when at least one scenario is identified for each query. The assertion is complete when all nine steps have been satisfied.

b) PTP 1: Forward Feed: What the Focal Complex Does

The forward feed derives the AIC structure outward from the focal complex toward its environment:

Step	Query Type	Question	Output
1	Scenario identification	What complexes and actions are in the scenario?	Scenario A (focal complex + interaction)
2	PrimeP definition	What is the supra-system's PrimeP? What is the goal?	Supra PrimeP + goal interaction
3	Auxiliary Influence	What is the complex trying to influence to manifest PrimeP?	Scenario B (influence)
4	Auxiliary Control	What should the complex control to enable the influence?	Scenario C (control)
5	Auxiliary Appreciation	What should the complex appreciate to enable the control?	Scenario D (appreciation)

The ordering A→B→C→D reflects the causal chain of AIC: purpose (A) is manifested through influence (B), which is enabled by control (C), which is enabled by appreciation (D). Each layer is a prerequisite for the one above it.

c) PTP 2: Backward Feed: What the Environment Does Back

The backward feed models the environmental response to the focal complex, manifesting its PrimeP, the reverberations that return:

Step	Query Type	Question	Output
6	Influencing Scenarios	What other scenarios, impacted by A's PrimeP, will influence A?	Scenario X
7	Controlling Scenarios	What other scenarios, impacted by A's PrimeP, will control A?	Scenario Y
8	Appreciative Scenarios	What other scenarios must appreciate A?	Scenario Z
9	Feedback to X, Y, Z	How does A's AIC feedback to X, Y, Z?	Bidirectional AIC map

This phase forces the analyst to model the scenario not as a one-way causal chain but as a closed loop: A acts on its environment → the environment responds → the response acts back on A.

d) Practitioner step-by-step implementation

STEP 0 Write your scenario (one paragraph)

PTP 1: FORWARD FEED (what your complex does):

Step 1: Name the focal complex + its key interaction (Scenario A)

Step 2: Name the supra-system + its PrimeP

Step 3: What does the focal complex INFLUENCE? (Scenario B , probabilistic)

Step 4: What does the focal complex CONTROL? (Scenario C , deterministic)

Step 5: What must the focal complex APPRECIATE? (Scenario D , sensing only)

PTP 2: BACKWARD FEED (what the environment does back):

Step 6: What INFLUENCES the focal complex from the environment? (Scenario X)

Step 7: What CONTROLS the focal complex from the environment? (Scenario Y)

Step 8: Who must APPRECIATE the focal complex? (Scenario Z)

Step 9: How does the focal complex FEED BACK to X, Y, Z?

CHECK: Rule D exhaustiveness: have all three AIC types appeared

in both the forward AND backward feeds?

If any type is missing → model is structurally incomplete.

e) The Full STCoT_2

Table F.9 AIC-Complexity Systemic predictive reasoner

Core STCoT	AIC-Complexity Systemic predictive reasoner
Input	Inquired Scenario or scenario
General Systems Model	General rule A: Ordered Relational Definition of a System (Smith): IF a complex S possesses a defined primary purpose P, AND there exists a set of entities and interactions $R = \{r_1, r_2, \dots, r_n\}$ such that each r_i either affects P or is affected by P, THEN S constitutes a system, defined as the ordered set of relationships between P and all members of R. Contrapositive: IF no defined purpose P can be attributed to S, OR no ordered relationships between S and its affecting/affected entities can be established, THEN S does not constitute a system under this definition. General rule B: Contextual Embeddedness of Purpose (Smith): IF a system S pursues a purpose P or goal G, THEN P or G is necessarily conducted within a

	context C, where C comprises both the system S itself and at least one external phenomenon E that is not fully contained within S. Corollary: IF a purpose or goal is modelled without reference to an external context, THEN the model is incomplete and may fail to account for interactions that affect or are affected by P. General rule C: Appreciative Awareness as a Behavioural Prerequisite (Smith): IF a system S exhibits purposeful behaviour directed toward achieving purpose P, THEN S must maintain appreciative awareness of both P itself (internal state) and the context C in which P is pursued (external environment), such that S can adjust its behaviour in response to changes in either P or C. Corollary: IF a system S acts toward P without appreciating its context C, THEN S's behaviour may produce unintended outcomes when C deviates from the assumed conditions under which S was designed or trained General rule D: AIC Relational Decomposition of Context (Smith): IF a system S operates within a context C in pursuit of purpose P, THEN C can be exhaustively decomposed into exactly three relational categories with respect to S: D1) Control: IF a complex or interaction X in C is directly governable by S (S can impose deterministic, intended changes on X), THEN X is classified as within S's sphere of control. D2) Influence: IF a complex or interaction X in C cannot be directly governed by S, BUT S can indirectly affect X through intermediary mechanisms with non-guaranteed probabilistic effect, THEN X is classified as within S's influence sphere. D3) Appreciation: IF a complex or interaction X in C can neither be controlled nor influenced by S, BUT X affects S's state, decisions, or performance, THEN X is classified as within S's appreciation sphere, and S must account for X without being able to govern it. Exhaustiveness condition: IF all entities and interactions in C have been classified under D1, D2, or D3, THEN the contextual relational decomposition of S with respect to P is structurally complete. Completeness: IF any complex X in C remains unclassified across D1, D2, and D3, THEN the Architect's model of S is incomplete and may harbour unmitigated epistemic uncertainty. General rule E: Recursive System Hierarchy (Mobus [1]): IF a complex S is classified as a system under Rule A, THEN: E1) Subsystem containment: S may contain one or more subsystems $\{S_1, S_2, \dots, S_k\}$, each of which satisfies Rule A independently with respect to its own purpose P_i, and each of which is bounded within S. E2) Supra system membership: S may itself be a subsystem of a larger supra system S_{sup}, which satisfies Rule A with respect to a higher-order purpose P_{sup} (Primary Purpose) that encompasses P. E3) Relational inheritance: IF S is a subsystem of S_{sup}, THEN the context C of S includes interactions originating from S_{sup} and its other subsystems, which must be accounted for under Rule D. LEADING TO: Predictable Systems Complexity
ArcPT	PTP 1: AIC Forward feed modelling Predictive queries (PQ) and guiding prompts (GP): 1) Inquired scenario: PQ: What complexes and actions are being noted in the Scenario? GP: identify a complex under enquiry and an interaction (Scenario A). 2) PrimeP/goal: PQ: What is the larger system (supra system) where the complex is part of, and what PrimeP does it serve? What is the noted Scenario and its gal interaction? GP: Define the supra source's PrimeP (supra source is the larger

	source complex where the complex is part of). In a Scenario, define the expected outcome (goal) of the AIC interactions. Auxiliary Influence: PQ: What is the complex trying to influence to enable or manifest the PrimeP? GP: Identify the influence interaction (Scenario B). Auxiliary Control: PQ: What should the complex control to enable or manifest the influence interaction? GP: Identify the control interaction (Scenario C). Auxiliary Appreciation: PQ: What should the complex appreciate to enable or manifest the control interaction? GP: Identify the appreciative interaction (Scenario D). PTP 2: AIC backward feed modelling The backwards feed models the environmental impact of the PrimeP on other Scenarios in the larger context, as feedback on the impact of those Scenarios on the PrimeP. Influencing Scenario: PQ: What other Scenarios whose PrimeP is directly or indirectly impacted by A manifesting its own PrimeP, are going to influence the Scenario under inquiry? GP: Identify the influences on the Scenario under inquiry (Scenario X). Controlling Scenario: PQ: What other Scenarios whose PrimeP is directly or indirectly impacted by A manifesting its own PrimeP, are going to control the Scenario under inquiry? GP: Identify the controls over the Scenario under inquiry (Scenario Y). Appreciative Scenario: PQ: What other Scenarios whose PrimeP is directly or indirectly impacted by A manifesting its own PrimeP, must appreciate Scenario under inquiry? GP: Identify the Scenario under inquiry (Scenario Z). AIC backward feedback to X, Y, Z: PQ: how does or should the Scenario under inquiry (a) feedback to X, Y,Z Scenarios? GP: define the AIC responses or impacts of Scenario A back to X,Y,Z Scenarios. <pre> graph LR subgraph Forward_Feeding [AIC forward feeding channels] A((Situation A)) -- "Aux Influence" --> B((Situation B)) A -- "Aux Control" --> C((Situation C)) A -- "Aux Appreciation" --> D((Situation D)) A -- "Primary Purpose/goal/outcome" --> D end subgraph Backward_Feeding [AIC backward feeding channels] Z((Situation Z)) -- "Appreciation" --> A X((Situation X)) -- "Influence" --> A Y((Situation Y)) -- "Control" --> A A -- "Situation A reactions to X,Y,Z" --> Z end </pre> Step completion criteria: The step is regarded as complete when at least one Scenario is identified for each query.
Output Prediction	Architect assertion: The Architect asserts that the predicted list of all AIC interactions is pragmatically consistent, structurally complete, and that their likelihood and impact are justified with respect to General Systems Model A, B, C, D, E.

Table F.10 AIC-Core STCoT Forward-feed implementation

Input	Step 1: Unsafe problematic activities: Adversarial drones follow train tracks, which may lead to striking the train [derived from n12].	
Step 2: Observed Complex (obs)	Step 3: Inquired interaction	Step 4: supra source Primary Purpose

roaming_adversarial_drone, train_derailment	Roaming adversarial drone approach train	Adversarial Scheme PrimeP: Disrupt Train Network operations.
Step 5: Auxiliary Influence interaction	Step 6: Auxiliary Control interaction	Step 7: Auxiliary Appreciation interaction
{roaming_adversarial_drone}_[-lead]_{train_derailment}	{roaming_adversarial_drone}_{+govern}_{roaming_adversarial_drone_flight_dynamics},{roaming_adversarial_drone}_[-approach]_{train_tracks_zone_fence_structure}	{roaming_adversarial_drone}_[-detect]_{visitble_train_tracks_zone_visible_structure}
Output prediction: Step 8: Predicted Problem Domain Contributing Scenarios (with repetition)		
The architect asserts that: Appreciation = ['roaming_adversarial_drone', 'visitble_train_tracks_zone_visible_structure'] Influence = ['roaming_adversarial_drone', 'train_derailment'] Control = ['roaming_adversarial_drone', 'roaming_adversarial_drone_flight_dynamics', 'roaming_adversarial_drone', 'train_tracks_zone_fence_structure']		

F.4 Hazards, Threats and Opportunities/Affordances Scenarios (HazTOPS)

Table F.11 Hazards, Threats, and Opportunities/Affordances Scenarios Structure

HazTOPS Aspect	Definition
Hazards, Threats or Opportunities/Affordances Scenario: Guide word	This section identifies Scenarios as hazardous or opportunistic with a "Guide word" headline. e.g., HazTOPS Operating Scenario: more conflict with Local People
Operating Scenario Context: What will happen? Articulate a possible Scenario.	This section provides a more detailed description of the Scenario. For example, people decide to deploy their drones to monitor the Agent, leading to a swarm of Local drones that distract the Agent.
Hazard, Threat or Opportunity definition (consider used-complexes): Why would it happen? Articulate the causes that may lead to such an eventuality.	What exactly is the source of danger or opportunity? Consider using complexes or tools to identify hazards or opportunities in a Scenario. e.g., an increased number of adversary drones piloted by local people in the vicinity.
Foreseeable Sequence	Specify the sequence of events that could occur due to a

of Events: How does it happen? Describe the sequence of Scenarios or events that could occur in the eventuality.	given Scenario. Visualise and predict the chain of events based on initial conditions. e.g., people utilise their back gardens adjacent to the Train Tracks Zone; People launch personal drones; Eagle Drone encounters additional adversarial drones.
Potential harm or benefit: So what? What would be of concern or interest? Predict the set of harms or affordances.	What is the ultimate outcome resulting from the danger or opportunity at hand? e.g., risk of drone collisions and decreased effectiveness of Eagle Drones.

Table F.12 HazTOPS guidewords partial definition for superficial predictive thinking

Guide words	Meaning
More: increased uncertainty due to the involvement of more things than initially expected.	Definition: Refers to increasing the number of interactions, elements, or complexities in a complex, which can lead to new opportunities or hazards. Example: The introduction of more adversary drones into the airspace increases the likelihood of conflict and signal interference.
Part of: increased uncertainty due to unexpected involvement of Part of the source or sink	Definition: Refers to the involvement of only a portion of a complex or interaction, which can disrupt the complex's overall effectiveness. Example: Only part of a Target Object of Interest (TOI) is salient, resulting in a false negative by the security drone.
Less effectiveness: increased uncertainty due to unexpected reduction of some expected effectiveness	Definition: Refers to the reduced effectiveness of a complex's actions or interactions, which may weaken its overall function and stability. Example: The Eagle Drone's sensors become less effective in fog, causing it to misidentify the train track boundaries.
Before or after, Late or early: increased uncertainty due to unexpected temporal, special or systematic sequence changes.	Definition: Refers to timing issues where actions or interactions occur earlier or later than intended, unexpected change in temporal sequence or special positioning, potentially resulting in failures or missed opportunities. Example: The Eagle Drone detects an adversary drone too late, missing the chance to control or neutralise it in time.

F.5 Stage 3A STCoTs

STCoT_3 is grounded in Alexander Bogdanov's Tektology [2]. Bogdanov's central insight was: Stability in any organised system arises not from the absence of opposing forces, but from their mutual neutralisation. Where antagonistic forces are absent, the system is volatile and susceptible to catastrophic transformation.

This principle is formalised in three General Systems Rules that govern STCoT_3:

Rule	Name	Core Claim
G	Equilibrium of Complexes Principle (Bogdanov)	Mutually antagonistic scenarios that neutralise each other produce systemic stability

H	Bidirectional Interaction Completeness Principle	Mutually antagonistic AIC interactions on shared complexes produce an equilibrium
I	Dialectical Equilibrium of Complexes Principle	A complex in dialectical AIC relationship with another, where AIC actions neutralise each other, is stable; absent reciprocal counterbalancing, the complex is volatile and susceptible to Black Swan failures

The practical implication is precise and actionable:

STABILITY requires:

FOR EACH problematic scenario P:

∃ counter-scenario C such that:

C.AIC_interactions NEUTRALISE P.AIC_interactions

at the same organisational level

INSTABILITY (Black Swan susceptibility) occurs when:

∃ problematic scenario P with NO adequate counter-scenario C

This is what STCoT_3 checks: whether the proposed solution produces a complete, balanced counter-interaction structure across the full AIC space of the problem.

F.5.1 The Three-Layer AIC Solution Structure

STCoT_3 uses the AIC framework not just to decompose problems, but to structure solutions.

For any problematic scenario requiring a response:

Problem scenario (target):

→ Requires INFLUENCE counter-interaction (primary stabiliser)

→ Enabled by CONTROL of the relevant problem component

→ Enabled by APPRECIATION of the context/state

Solution AIC structure (counter-interaction):

Influence counter ← enabled by Control ← enabled by Appreciation

This is the forward-feed AIC solution chain that STCoT_3 maps. It ensures that the solution is not a single-point intervention but a complete three-layer coupled response that maintains equilibrium at the influence, control, and appreciation levels simultaneously.

F.5.2 The STCoT_3 Structure: Five-Step Equilibrium Mapping Process

STCoT_3 PROCESS

INPUTS:

1. Architect high-level solution prescriptions
2. Engineering problem space AIC model

GSM ANCHOR: Rules G (Bogdanov), H (Bidirectional), I (Dialectical)

STEP 1: Identify the PRIMARY PROBLEMATIC SCENARIO

→ What requires counterbalancing to achieve equilibrium?

STEP 2: Define the COUNTER-INFLUENCE interaction

→ What interaction does the solution exert to neutralise?

STEP 3: Define the CONTROL interaction

→ What must the solution CONTROL to enable the influence?

STEP 4: Define the APPRECIATION interaction

→ What must the solution APPRECIATE to enable the control?

STEP 5: CHECK equilibrium completeness

→ Does every problem scenario have an AIC counter?

→ Are antagonistic interactions present at all three

AIC levels (Control, Influence, Appreciation)?

OUTPUT: Architect Assertion , complete, balanced AIC solution map

grounded in Rules G, H, I

Completion criterion: The step is complete when all forward-feed AIC feedback loops between the autonomous system (solution) and the problematic scenario have been modelled. Every problem-side AIC interaction must have at least one solution-side counter-interaction at the same organisational level.

F.5.3 Practitioner step-by-step implementation

STEP 0 Gather inputs:

(a) Architect solution prescriptions

(b) Problem space AIC model (from STCoT_1 or STCoT_2)

STEP 1 Map the PROBLEM SPACE:

For each destabilising complex, define its AIC interaction:

→ AIC type (Control / Influence / Appreciation)

→ Source → Sink

→ Action and destabilising effect

STEP 2 Define COUNTER-INFLUENCE interaction:

"What does the solution INFLUENCE to neutralise the primary problem?"

STEP 3 Define CONTROL interaction:

"What must the solution CONTROL to enable that influence?"

STEP 4 Define APPRECIATION interaction:

"What must the solution APPRECIATE to enable that control?"

STEP 5 Run the EQUILIBRIUM TEST (Rules G, H, I):

For each problem interaction P:

→ Does a solution counter-interaction exist at the SAME AIC type?

SAME TYPE → BALANCED

DIFFERENT TYPE → PARTIAL (flag residual risk)

NONE → UNBALANCED (Black Swan vulnerability)

Rule H: solution must have all three AIC types present

Rule I: every problem AIC must have a counter-AIC

F.5.4 STCoT_3 Solution-Problem Coupling Systemic predictive reasoner

Table F.13 Solution-Problem Coupling Systemic predictive reasoner

STCoT_3	Solution-Problem Coupling Systemic predictive reasoner
Input	Architect high-level solution prescriptions tables. Engineering problem space AIC models
General Systems Model	General rule G: Equilibrium of Complexes Principle: IF a complex contains mutually antagonistic Scenarios (A and B) that neutralise each other's capability in a given OrgXP THEN the OrgXP remain in equilibrium (systemic stability/predictability) unless perturbed through changes in internal or external uncertainty. [Bogdanov [2]] General rule H: Bidirectional Interaction Completeness Principle (AIC Interpretation): IF a complex contains mutually antagonistic AIC interactions acting on shared complexes, that neutralise each other's capability in a given organisational experience, THEN the complex exists in equilibrium (systemic stability/predictability). General rule I: Dialectical Equilibrium of Complexes Principle (AIC Interpretation) IF complex X exists in a dialectical AIC relationship with complex Y, comprising mutually antagonistic control, influence, and appreciation interactions acting on shared complexes such that the AIC actions of X neutralise the AIC actions of Y, and vice versa, at a given organisational level, THEN complex X exists in equilibrium (systemic stability/predictability). Complexes lacking such reciprocal counterbalancing AIC interactions are unstable (volatile) and susceptible to cascading Black Swan failures. Figure 4.7 model manifests G, H, and I rules by modelling the mutually antagonistic interaction groups comprising control, influence, and appreciation dynamics that counteract one another in a given organisational scenario, thereby neutralising net systemic effects and maintaining interaction stability (predictability) within the scenario's complexity. Complexes lacking reciprocal counterbalancing interactions are unstable (volatile) and susceptible to cascading Black Swan failures. The following AIC general solution space model is consistent with the General Systems Model G,H,I. The diagram illustrates the AIC solution space model. It features a central red oval labeled 'OrgXP Equilibrium'. To its left is a green oval labeled 'Problematic Situation', and to its right is a blue oval labeled 'Antagonistic Solution'. Three curved arrows connect the central node to the left node: a green arrow labeled 'Aux Appreciation', a red arrow labeled 'Aux Influence', and a blue arrow labeled 'Aux Control'. Three curved arrows connect the central node to the right node: a green arrow labeled 'Aux AIC', a red arrow labeled 'Aux AIC', and a blue arrow labeled 'Aux AIC'. Above the diagram, the text 'Forward-feed Effect' is written. Below the diagram, the text 'Aux AIC' is written. LEADING TO: Asserted Organisational equilibrium. Which means the likelihood of organisational change approaches zero because opposing forces cancel each other's transformative effects (impact) on the system [Bogdanov]. Figure F.2 AIC solution space model

Appendix F

ArcPT	Predictive query 6.1: What is the primary problematic Scenario requiring influence (counterbalancing) to achieve systemic equilibrium? What Scenario requires control to enable the counterbalancing activity? What Scenario requires appreciation to ensure that the autonomous system maintains adequate control? Guiding prompt 6.2: Define the autonomous systems Forward-Feed AIC interactions with the rest of the problem. Map the autonomous systems to all the parts of the problematic Scenario and define the actions and impact types for each interaction. STEP 0 Gather inputs: (a) Architect solution prescriptions (b) Problem space AIC model (from STCoT_1 or STCoT_2) STEP 1 Map the PROBLEM SPACE: For each destabilising complex, define its AIC interaction: → AIC type (Control / Influence / Appreciation) → Source → Sink → Action and destabilising effect STEP 2 Define COUNTER-INFLUENCE interaction: "What does the solution INFLUENCE to neutralise the primary problem?" STEP 3 Define CONTROL interaction: "What must the solution CONTROL to enable that influence?" STEP 4 Define APPRECIATION interaction: "What must the solution APPRECIATE to enable that control?" STEP 5 Run the EQUILIBRIUM TEST (Rules G, H, I): For each problem interaction P: → Does a solution counter-interaction exist at the SAME AIC type? - SAME TYPE → BALANCED - DIFFERENT TYPE → PARTIAL (flag residual risk) - NONE → UNBALANCED (Black Swan vulnerability) Rule H: solution must have all three AIC types present Rule I: every problem AIC must have a counter-AIC Step completion criteria 6.3: The step is considered complete when all forward-feed AIC feedback loops between the autonomous system and the problematic Scenario have been modelled.
Output Prediction	Architect assertion 6.3: the Architect asserts that the output predictions are pragmatically consistent, structurally complete, and they are acceptably likely and impactful, with respect to the validity and applicability of general systems rules G,H,I: Figure 4.8 Figure F.3 Active_avp AIC solution space model example

Table F.14 AIC Interactions capture for system-level requirements

Source: source complex for example {eagle_drone}	
Output Behaviour	Input Behaviour that impacts the emergence of Output Behaviour
In: influence interaction	An: Appreciative interactions.
	Cn: Control interactions
I2: {flying_eagle_drone}_[-inhibit]_{roaming_adversarial_drone}	A1: {flying_eagle_drone}_[-recognise]_{visible_vegetation_visual_appearance}

	A3: {flying_eagle_drone}_[-recognise]_{visible_adv_drone_visual_appearance}
	C1: {flying_eagle_drone}_[-inhibit]_{roaming_adversarial_drone_flight_dynamics} C2: {flying_eagle_drone}_[-avoid]_{visible_vegetation_structure}

F.6 Multi-Purpose Operational Design Domain Definition (ODD)

The definition of an Operational Design Domain (ODD) is a critical step in the systems engineering lifecycle for safety-critical autonomous systems. The ODD specifies the conditions under which the autonomous system is expected to operate safely and effectively, serving as the foundation for system design, validation, and operational assurance. Below, we outline a generic approach for defining an ODD, followed by an example-based explanation and a structured table to formalise the definition. The architect can decide on the granularity of the Definition. However, we recommend that safety-critical applications use all the categories mentioned in Appendix A. Appendix B defines the operational environment uncertainty grade.

F.6.1 Step 1: Identify the System of Interest

The first step is defining the autonomous system the ODD applies. This includes understanding the purpose and context of its deployment. For example, suppose the system is an autonomous drone like the Eagle Robot. In that case, the focus might be on surveillance and operational support within a specific environment, such as a train track zone.

F.6.2 Step 2: Define the Solution Operational Space

The operational space refers to the physical and functional domain where the system will be deployed. This step involves identifying the geographical region, infrastructure, and the boundaries of the system's operational zone. For instance, the operational space for the Eagle Robot could be "Train Track Zone in the UK," where specific challenges such as restricted airspace and proximity to critical infrastructure are considered.

F.6.3 Step 3: Determine the ODD Uncertainty Grade

Assign an uncertainty grade to represent the complexity and variability of the operational environment. Safety-critical applications typically range from Grade A (ideal and highly controlled conditions) to Grade E (highest uncertainty conditions). For example, in a Grade A Scenario, the Eagle Robot might operate in clear weather conditions with minimal environmental variability.

F.6.4 Step 4: Specify Development Purpose

Articulate the intent behind defining the ODD, such as system training, testing, or operational deployment. For the Eagle Robot, the ODD may be designed to train the system for obstacle detection and test its stability under specific environmental conditions.

F.6.5 Step 5: Characterise the Environment

Define the environmental parameters that the system is designed to handle, including weather, lighting, and natural phenomena. This step ensures the system is only exposed to conditions it has been validated to handle. For instance, for the Eagle Robot, the environment might include:

- Lighting.
- Weather.
- Time of the Year: Seasons-specific environmental characteristics.
- Landscapes type variety definition.
- Geographical region-specific natural phenomena.
- Time of the Day.
- Perceived Horizon Attitude.
- Sun sphere positioning.
- Moon sphere positioning.
- Specialised zones features.

F.6.6 Step 6: Formulate the ODD in a Structured Table

Consolidate the ODD definition into a structured table that provides clarity and traceability for all stakeholders. The table should include:

- **System of Interest:** The autonomous system being defined.
- **Solution Operational Space:** The geographical and functional boundaries.
- **ODD Uncertainty Grade:** The level of complexity in the operational environment.
- **Development Purpose:** The primary intent of defining the ODD.

- **Environmental Characteristics:** Detailed environmental conditions, such as lighting, weather, and visibility.

Table F.15 Example ODD format

System of interest		Eagle Robot
Solution Operational Space		Train Track Zone in the UK
ODD Uncertainty Grade		Grade A
Development purpose		Training and Testing
Environment Characteristics		Ideal natural environment system for perception
Natural Lighting Conditions		Sunny
Weather Conditions	Precipitation mm/h	0
	Wind km/h	<5
	Humidity %	50
	Visibility km	>8
	Cloud Cover	Clear or Blue Sky (0/8 oktas)
	Snow mm/12 hrs	0
	Pollen	No pollen
	Sand	Neither sand nor dust storms
	Temperature	15-25 degrees Celsius
	Sunshine Duration	12 hours
Time of the Year: Seasons-specific environmental characteristics		1 type of season
Landscapes type variety definition		1 type of Landscapes
Geographical region-specific natural phenomena		0 or 1 type of Infrastructure
Time of the Day		Midday (Noon)
Perceived Horizon Attitude		1 type
Sun sphere positioning		1
Moon sphere positioning		0
Specialised zones features		1 feature

F.7 Stage 4: AIC perspective shift

AIC perspective shift is a lateral thinking technique for Black Swan Scenarios discovery. A Black Swan Scenario is a lesser expected, atypical, least salient, whose impact cannot be ignored Scenario in the real world. However, an intelligent system is expected to make the right decisions,

for example, the food delivery robot in a Scenario like the one described above. In the literature, the term “Edge Cases” [3] is commonly used to describe the harder-to-predict Scenarios. However, we believe such characterisation is an inappropriate adaptation of a concept well understood in deterministic software development, where maximum and minimum input boundaries are identifiable. For non-deterministic complexes such as ML-based perception, there is no recognisable boundary of inputs for pictorial datasets. For example, how can we specify a “human” in a pictorial sense to clearly define maximum or minimum edge cases? Using the term “Black Swans, which are Hard-to-Foresee Emergent Scenarios”, is more appropriate, for example, when facial recognition datasets did not account for people wearing COVID-19 protective masks.

Predicting Black Swan Scenarios is crucial for robust design and operational resilience in complicated architecture.³ Common hazards and risk analysis techniques may help developers discover unsafe Scenarios. However, more needs to be done regarding intelligent systems design since their emergence is exponentially more unpredictable than that of non-intelligent systems. The AIC (Appreciation, Influence, Control) perspective shift technique, as applied within the AIC General Systems approach (GSApp), enables architects to predict potential harder-to-predict unsafe Scenarios by shifting perspectives within known events. This approach triggers the architect’s predictive thinking by reimagining events from different angles within the AIC framework.

F.7.1 Predictive atypical AIC Perspective Shifts Approaches

The AIC perspective shift is a lateral Predictive Thinking Approach that leverages a dual-dimensional approach within the AIC framework, facilitating a re-evaluation of known events or interactions from alternative perspectives. This technique comprises four types of shifts: altering the AIC type of an interaction, reversing the direction of the interaction, altering the type of impact for an action and altering all kinds together.

The term “Disordered” relates to the fact that by the end of the requirements modelling of the complex, the architect would have generated an AIC-ordered model (assumed AIC intuitive model) of the complexity field. In this step, we challenge the architect's imagination (predictive capability) to consider Scenarios by altering the underlying perspective (considering the potential of a complementary, confusing, complex or counterintuitive model). This allows for discovering Black Swan Scenarios (concerning the complexity field of the solution) or black-sawn events. This

³ By “resilience,” we mean the robustness of the solution architecture to cope with the change in the complexity field, thus remaining relevant and effective in delivering the desired PrimeP. In simple terms, resistance to change of requirements.

approach is an objective approach for lateral thinking since it takes the thought approach out from what is deemed to be intuitive into a counter-intuitive perspective, thus supporting the architect in utilising lateral thinking.

f) Predictive Perspective Shift by Altering Interaction's AIC Type

This dimension entails modifying the AIC type of interaction within a complex, transitioning an element's role from appreciation to influence or control or vice versa. This allows architects to explore how changes like interaction affect complicated behaviour and outcomes. The shift involves redefining the functional purpose of an element in a Scenario, such as changing an advisory tool into an authoritative controller. This transformation prompts a re-evaluation of complicated dynamics and user interaction patterns.

Example

Consider an aircraft avoidance recommendation complex (source) initially designed to influence the pilot's decision-making approach (sink) in a supportive manner. Applying the perspective shift, the complex's interaction is reimagined as controlling, potentially obstructing the effectiveness of the pilot's intuition role in the autonomous decision-making approach.

Black Swan Scenario prediction

- **Initial Scenario (Influence):** The relationship between the recommendation complex and the pilot is primarily influential and supportive. Pilots might initially perceive the complex as somewhat unreliable, preferring to rely on their intuition.

The perspective is then can be shifted from influence to control to view this complex as a controlling complex:

- **Post-shift perspective Scenario (Control):** If we alter the perspective, we can envision the recommendation complex directly controlling the pilot's behaviours, particularly in high-stress Scenarios where the pilot may depend entirely on the complex's directives, potentially to the detriment of safe flying practices. Such a Scenario can be considered as a surprising, Black Swan Scenario.

g) Predictive Perspective Shift by Reversing Interaction Direction

This approach reverses the flow of interaction, transforming the source of an action into its recipient (sink), and vice versa. This reversal challenges conventional flow dynamics within the complex, prompting new insights into potential complicated behaviours and vulnerabilities. Reversing the interaction direction can reveal vulnerabilities or strengths in the complex by showing how changes in the flow of information or control can lead to different outcomes.

Example

Weather conditions (source) influence an aircraft collision avoidance recommendation complex (sink). The complex must appreciate the weather when attempting to control the types of recommendations it makes while influencing the pilot's decision-making approach to avoid a potential collision.

Black Swan Scenario prediction

- **Initial Scenario (Influence):** An initial interaction between the weather conditions (appreciated sink) and an aircraft collision avoidance recommendation complex (appreciating source), with the recommendation complex being influenced by environmental conditions.

By reversing this interaction, the aircraft can be imagined as an influencer of the environment:

- **Post-shift perspective Scenario (reversed influence):** If we alter the influence direction, we have a Scenario where the aircraft influences the environment. This would be counterintuitive and surprising; what kind of Scenario could the aircraft be viewed as having influenced some aspect of the environment? This might involve the aircraft's complexes modifying environmental data inputs due to a cyber-attack or a bug, which could lead to erroneous sensor data readings being fed into navigational or operational complexes. In the context of Eagle Robot (Figure F.4):

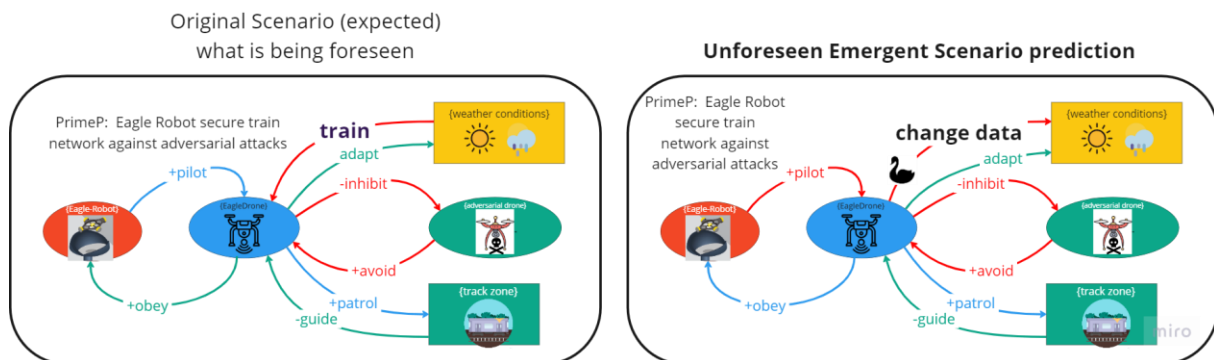


Figure F.4 Reverse interaction perspective shift model

h) Predictive Perspective Shifts by altering types of impact [Supportive, Obstructive, Neutral]

The perspective shift by altering the type of impact involves changing the impact that the actions of one complex (source) have on another complex (sink) in terms of being supportive, obstructive, or neutral. This shift allows architects to evaluate the adaptability and resilience of complexes under different operational conditions by envisioning how variations in the impact type could alter behaviours and outcomes. This analysis is crucial for understanding a complex's potential vulnerabilities or strengths that may not be evident under normal conditions.

Example

In its standard operational mode, an aircraft collision avoidance complex enhances safety by providing timely warnings and manoeuvre suggestions to avoid potential aerial conflicts. The complex continuously monitors the airspace for other aircraft and uses algorithms to predict potential collision paths. It then alerts pilots with specific avoidance manoeuvres.

Black Swan Scenario prediction

- **Initial Scenario (supportive effect):** The complex supports the pilot by augmenting their Situational awareness and decision-making capabilities, helping to maintain safe distances from other aircraft and navigate congested airspaces effectively.
- **Post-shift perspective Scenario (alternative effects):** The perspective is then can be shifted by re-imagining the type of effects, changing from supportive to obstructive or neutral:

Obstructive Effect

1. **Post-shift perspective:** By altering the perspective to an obstructive effect, we can reimagine the collision avoidance complex as temporarily hindering the pilot's ability to execute certain manoeuvres.
2. **Operational Scenario:** During a high-density traffic Scenario at an airport, the complex might excessively restrict pilot manoeuvre options to prevent potential collisions. This might force pilots to follow less optimal paths or delay landings, which could disrupt flight schedules and increase fuel consumption.
3. **Impact:** Here, the complex, usually supportive, now acts as an obstruction to efficient flying, prioritising safety over efficiency and convenience.

Neutral Effect

1. **Post-shift perspective:** Shifting to a neutral effect, the complex could be imagined as non-responsive or minimally interactive, neither aiding nor hindering the pilot actively.
2. **Operational Scenario:** Consider a Scenario where the complex enters a diagnostic mode during flight, ceases to provide active collision warnings, but continues monitoring airspace without issuing commands.
3. **Impact:** In this mode, the complex neither supports nor obstructs the pilot, leaving them to rely entirely on their training and other navigational aids. This could be particularly challenging during low visibility conditions or in highly trafficked airspaces.

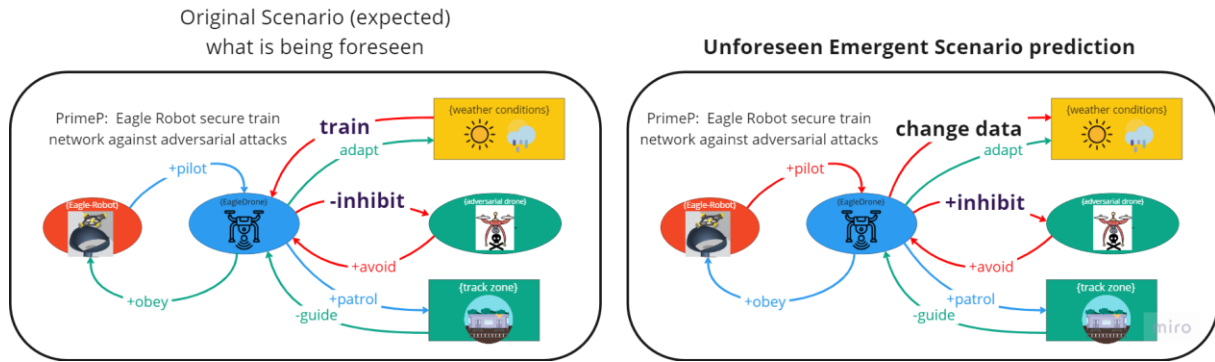


Figure F.5 Altering types of impact perspective shift model

i) Predictive Multi-types Shifts by Altering AIC and Impact Types, as well as Interaction Direction

This comprehensive shift involves altering the AIC type and impact type, and reversing the direction of interaction, providing a robust approach for evaluating complex interactions. This multidimensional shift enables exploration of complex Scenarios where changes in operational dynamics might occur. By simultaneously shifting the AIC type and interaction direction, architects can simulate complicated interactions that might not be apparent under normal analysis conditions.

Example

Initially, the aircraft avoidance recommendation complex influences pilot decision-making:

- **Initial Scenario:** The complex provides recommendations, and the pilot uses these to make informed decisions.

The Scenario is then reimagined with both the AIC type and interaction direction altered:

- **Post-shift perspective Scenario:** The pilot controls the recommendation complex's sensitivity settings, directly manipulating its operational parameters and recommendations. This thought approach may prompt us to question what kind of Scenario in which the pilot side controls the recommendation parameters. Does this mean we need to give the pilot such a level of control? Also, it may prompt us to question whether there may be a cyber-attack that exploits pilot access to the recommendation complex.

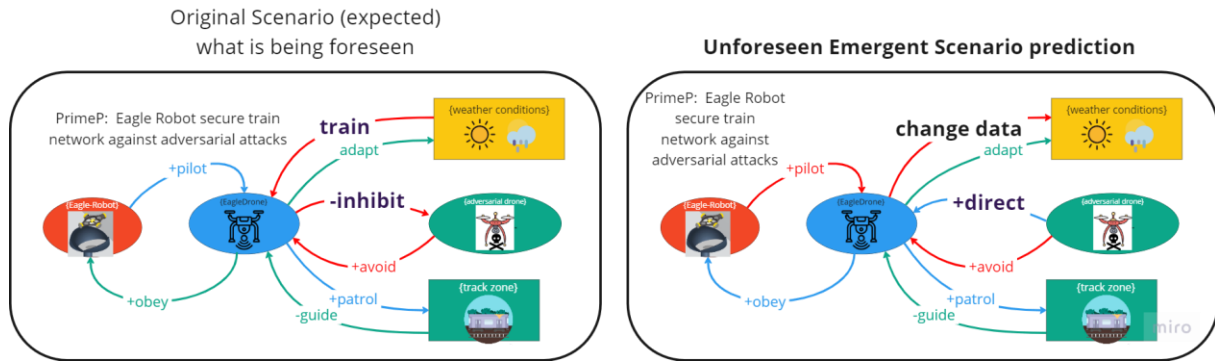


Figure F.6 Multi-types Shifts model

F.7.2 STCoT_4 AIC perspective shift-based unforeseeability systemic predictive reasoner

Before STCoT_4 can operate, the analyst must have produced an AIC complexity model a structured representation of a scenario using the three relational categories introduced in STCoT 3. An AIC interaction between a source complex S and a sink complex K is characterised by three semantic attributes:

Interaction := (Goal_Type G, Flow_Direction D, Impact_Polarity P)

$G \in \{\text{Appreciation, Influence, Control}\}$

$D \in \{S \rightarrow K, K \rightarrow S\}$

$P \in \{\text{Supportive, Obstructive, Neutral}\}$

An equilibrium model is an AIC model in which all interactions are balanced; for every destabilising interaction, a neutralising counter-interaction exists at the same organisational level. This is the model the analyst believes is representative of the current complexity.

STCoT_4's theoretical innovation is this:

If the equilibrium model is accurate, then any unmodelled interaction that disrupts that equilibrium is, by definition, a candidate Black Swan scenario, precisely because it was not represented in the analyst's model.

The General Systems Model provides two governing rules:

Rule J (Black Swan Emergence Principle): If an atypical AIC interaction emerges into a balanced model, AND no counter-interaction neutralises it, THEN equilibrium is disrupted $\rightarrow$ novel, structurally unstable OrgXP $\rightarrow$ Black Swan conditions.

Rule K (Atypical Disruption Principle): If any *semantic attribute* of an existing AIC interaction is altered, AND no antagonistic counter-interaction neutralises the change, THEN the result is an

atypical disruption , a candidate Black Swan , whose causal pathway is *non-salient under the analyst's normal ATD*.

The power of Rule K is profound: it says **you do not need to imagine entirely new scenarios**. You only need to **systematically alter the semantic attributes of existing interactions** , and the resulting scenarios, being structurally outside the equilibrium model, are *by definition* attentionally invisible to the analyst who built that model.

This is the formal basis for STCoT_4's four shift operations.

Each shift type applies a specific semantic perturbation to a source-sink interaction pair from the equilibrium model. They are ordered by the number of attributes simultaneously altered:

- Shift Type 1: Alter G only (1 attribute changed)
- Shift Type 2: Alter D only (1 attribute changed)
- Shift Type 3: Alter P only (1 attribute changed)
- Shift Type 4: Alter G + D + P (all 3 attributes changed simultaneously)

Shift Type 4 produces scenarios of maximum structural novelty; every semantic attribute of the original interaction is simultaneously transformed, producing a scenario that is structurally orthogonal to the equilibrium model and maximally attentionally invisible.

INPUT: An AIC equilibrium model with ≥ 1 source-sink interaction pair

FOR EACH source-sink pair (S, K) with interaction (G, D, P):

Step 1 (Shift Type 1):

- Alter G to $G' \neq G$ (hold D, P constant)
- Ask: what scenario emerges under goal type G' ?
- Completion: articulate a structurally distinct candidate Black Swan

Step 2 (Shift Type 2):

- Reverse D: $S \rightarrow K$ becomes $K \rightarrow S$ (hold G, P constant)
- Ask: what scenario emerges when roles are inverted?
- Completion: articulate a structurally distinct candidate Black Swan

Step 3 (Shift Type 3):

- Alter P to $P' \neq P$ (hold G, D constant)
- Ask: what scenario emerges when effect polarity inverts?
- Completion: articulate a structurally distinct candidate Black Swan

Step 4 (Shift Type 4):

- Simultaneously alter $G \rightarrow G'$, reverse D, alter $P \rightarrow P'$

- Ask: what maximally counter-intuitive scenario emerges?
- Completion: articulate a maximally novel candidate Black Swan

OUTPUT: 4 candidate Black Swan scenarios per source-sink pair,
each grounded in a formal General Systems Rule (J or K),
each structurally distinct from the equilibrium model.

Here is the canonical worked example traced through all four shifts:

Equilibrium interaction: Collision Avoidance System (CAS) → Pilot

- G = Influence (advisory, non-binding)
- D = CAS → Pilot
- P = Supportive (enhances pilot decision quality)

Shift	Alteration	Candidate Black Swan
Type 1	G: Influence → Control	CAS now issues binding directives. Pilot becomes operationally dependent on CAS; pilot intuition suppressed at safety-critical moments.
Type 2	D: CAS→Pilot reversed to Pilot→CAS	Pilot inadvertently/maliciously configures CAS sensitivity. Software feedback loop corrupts sensor inputs. Silent safety-critical blind spot.
Type 3	P: Supportive → Obstructive	Same control action, same direction, but now CAS issues excessive restriction commands, constraining pilot's manoeuvre options, forcing suboptimal routing.
Type 4	G: Influence→Control, D reversed, P: Supportive→Obstructive	Pilot controls CAS configuration (reversed), imposing binding constraints (Control), that obstruct CAS's ability to generate reliable recommendations. Cyber-attack or social engineering exploits this pathway silently.

j) How practitioners Apply STCoT 4

STEP 0 Build your AIC equilibrium model using STCoT
(identify all source-sink pairs and their G, D, P attributes)

STEP 1 Select one interaction pair (S, K) with attributes (G, D, P)

STEP 2 Apply Shift 1: change G → G' , what new scenario emerges?

STEP 3 Apply Shift 2: reverse D , what new scenario emerges?

STEP 4 Apply Shift 3: change P → P' , what new scenario emerges?

STEP 5 Apply Shift 4: change G+D+P simultaneously , what new scenario?

STEP 6 Check Architect Assertion 7.13:

- (i) Consistent with Rules J and K?
- (ii) Structurally distinct from equilibrium and each other?
- (iii) Acceptably likely/impactful?

(iv) One scenario per shift type?

STEP 7 Document all four as candidate Black Swan scenarios

STEP 8 Repeat for every source-sink pair in the model

k) Full STCoT 4

Table F.16 AIC perspective shift-based unforeseeability systemic predictive reasoner

STCoT_4	AIC perspective shift-based unforeseeability systemic predictive reasoner
STCoT Input	A typical AIC Complexity model of some Scenarios
General Systems Model	General rule J: Black Swan Emergence Principle:[Derived from Rule I: Dialectical Equilibrium of Complexes Principle] IF an atypical and original AIC interaction emerges into an AIC-balanced model (that is, an interaction whose source complex, sink complex, or interaction character (goal type, action type, impact polarity) was not present in the original equilibrium model) AND no antagonistic AIC counter-interaction exists within the model to neutralise its impact at the given organisational level, THEN the equilibrium of the affected complex is disrupted, producing an novel OrgXP that is structurally unstable and mayor may not be desirable. LEADING TO an atypical AIC complexity in which Black Swan scenarios are more likely to occur, which may or may not be hazardous. General rule K: Atypical Disruption Principle: IF a pre-determined AIC model of scenario complexity is in equilibrium, AND any semantic attribute of one or more of its AIC interactions is altered (i.e., goal type, action type, flow direction, or impact polarity changes), AND the altered interaction has no corresponding antagonistic counter-interaction within the model at the same organisational level, THEN the resulting interaction state constitutes an atypical and possibly original disruption of the model's equilibrium (a candidate Black Swan Scenario) whose causal pathway is non-salient under the Architect's normal Attention Distribution (ATD). LEADING TO an atypical AIC complexity in which Black Swan scenarios are more likely to occur, which may or may not be hazardous.
ArcPT	Shift Type 1: AIC Goal Type Disruption Principle IF an AIC-driven OrgXP is in equilibrium, such that a source complex pursues a defined goal of type $G \in \{\text{Appreciation, Influence, Control}\}$ with respect to a sink complex, producing a stable, expected interaction character within the model; AND the goal type G is altered to $G' \in \{\text{Appreciation, Influence, Control}\}$, $G' \neq G$, such that the source complex now pursues a qualitatively different relational purpose with respect to the sink complex (for example, an Influence goal is recharacterised as a Control goal, or an Appreciation goal is recharacterised as an Influence goal); AND no antagonistic counter-interaction exists within the current equilibrium model at the same organisational level to neutralise the impact of the altered goal type; THEN the equilibrium of the affected OrgXP is disrupted, the source complex's relationship to the sink complex is recharacterised in a manner

that was non-salient under the Architect's normal ATD, producing a novel, structurally unstable OrgXP that the Architect's clockwise-reasoning model did not predict;

LEADING TO a Black Swan scenario which may or may not be hazardous, in which the source complex exercises a qualitatively different form of influence over the sink complex than the equilibrium model assumed, potentially producing emergent dependencies, capability suppressions, or behavioural takeovers that constitute candidate Black Swan Scenarios.

Instantiating Predictive Enquiry and Guiding Prompt, STCoT_4 Step 1

General Systems Rule basis: Given a source or sink complex, it is possible that over time and with a change of complexity, the AIC goal type of the interaction is altered, leading to a new Scenario of complicatedness that was not present in the original equilibrium model.

Predictive Query 7.1:

What would happen if the interaction flow direction and impact polarity remained the same, but the goal's AIC type and corresponding action type were altered, specifically, if what was previously an Appreciation or Influence goal became a Control goal, or vice versa?

Guiding Prompt 7.2:

Review the AIC interaction of the identified source–sink pair in the equilibrium model. Identify the current goal type (Appreciation, Influence, or Control). Alter the goal type to one of the two remaining types and redefine an appropriate action that instantiates the new goal type whilst preserving the original flow direction and impact polarity. Describe the operational Scenario that would emerge under this altered goal type. Consider whether atypical AIC timing (§E.2.4), for example, Control preceding Appreciation, is implicated by the shift. Ask: what would it mean for this complex to exercise this new form of relational purpose over the sink? What emergent dependencies or behavioural constraints would be introduced?

Completion Criterion 7.3:

The step is considered complete when a Scenario has been articulated in which the source complex demonstrably exercises a qualitatively different AIC goal type over the sink complex, and the resulting operational scenario is recognisably distinct from the equilibrium model, constituting a candidate Black Swan Scenario grounded in the AIC Goal Type Disruption Principle.

Illustrative Example: An aircraft collision avoidance recommendation complex (source) initially pursues an Influence goal with respect to the pilot (sink): it provides advisory recommendations that the pilot may accept or override. The Influence goal is characterised by an indirect, non-binding effect, the pilot retains autonomous decision authority. Applying Shift Type 1, the goal type is altered from Influence to Control. The recommendation complex is now reimagined as directly controlling the pilot's decision-

	making in high-stress Scenarios, issuing binding directives that the pilot cannot override. This shift produces a candidate Black Swan OrgXP: a Scenario in which the pilot becomes operationally dependent on the complex's directives, potentially suppressing the exercise of pilot intuition at precisely the moments when intuition is most safety-critical. The equilibrium model did not anticipate this dependency because the Influence goal type was assumed to be stable. Shift Type 2: Reverse the Interaction Flow Direction Shift Type 2: Interaction Direction Reversal Principle IF an AIC-driven OrgXP is in equilibrium, such that a source complex S exerts an interaction of goal type G and impact polarity E upon a sink complex K, establishing a directed, stable interaction flow $S \rightarrow K$ within the model; AND the interaction flow direction is reversed, such that the former sink complex K becomes the source of a new interaction directed toward the former source complex S, producing the reversed flow $K \rightarrow S$, whilst the goal type G and impact polarity E are held constant; AND no antagonistic counter-interaction exists within the current equilibrium model at the same organisational level to neutralise the impact of the reversed flow; THEN the equilibrium of the affected OrgXP is disrupted, the relational roles of source and sink are inverted in a manner that was non-salient under the Architect's normal ATD, producing a novel OrgXP in which the original sink complex now exercises the original interaction type upon the original source complex; LEADING TO a Black Swan scenario which may or may not be hazardous , in which the operational roles of agent and environment, controller and controlled, or appreciator and appreciated are reversed, potentially producing emergent vulnerabilities, exploitation pathways, or feedback loops that constitute candidate Black Swan Scenarios. Instantiating Predictive Enquiry and Guiding Prompt: STCoT_4 Step 2 <i>General Systems Rule basis:</i> Given a source or sink complex, it is possible that over time and with a change of complexity, the direction of interaction is reversed, leading to a different Scenario of complicatedness in which the original recipient of an interaction becomes its originator. Predictive Query 7.4: What would happen if the interaction goal type and action type remained the same, but the interaction flow direction were reversed, such that the original sink complex became the source of the interaction, and the original source complex became its recipient? Guiding Prompt 7.5: Review the identified source–sink pair and the direction of their interaction in the equilibrium model. Swap the source and sink roles: the complex that previously received the interaction now initiates it, and the complex that
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	previously initiated it now receives it. Preserve the original goal type and impact polarity in the reversed interaction. Define an appropriate action that instantiates the reversed flow. Describe the operational Scenario that would emerge. Consider whether atypical AIC timing (§E.2.4), for example, the former sink acting before the former source has completed its appreciation cycle, is implicated. Ask: What does it mean for this formerly passive complex to now be the agent of this interaction? What vulnerabilities, exploitation pathways, or emergent feedback dynamics does this reversal expose? Completion Criterion 7.6: The step is considered complete when a Scenario has been articulated in which the original flow direction is clearly reversed, the former sink complex is demonstrably acting upon the former source complex, and the resulting operational scenario constitutes a candidate Black Swan Scenario grounded in the Interaction Direction Reversal Principle. Illustrative Example: Weather conditions (source) exert an Influence interaction upon an aircraft collision avoidance recommendation complex (sink): the complex must appreciate weather data when formulating avoidance recommendations. The equilibrium flow is: weather → complex. Applying Shift Type 2, the flow is reversed: the complex is now reimagined as influencing the environmental data inputs, the former sink is now the source. This reversal produces a candidate Black Swan OrgXP: a Scenario in which the recommendation complex's outputs, due to a software fault, sensor feedback loop, or cyber-attack, corrupt the environmental sensor data being fed back into navigational systems, producing erroneous Situational awareness across the broader avionics architecture. The equilibrium model did not anticipate this reversal because the weather was assumed to be an unalterable environmental source Shift Type 3: Alter the Interaction Impact Type Shift Type 3: Impact Polarity Disruption Principle IF an AIC-driven OrgXP is in equilibrium, such that a source complex exerts an interaction of goal type G upon a sink complex with impact polarity $P \in \{\text{Supportive, Obstructive, Neutral}\}$, producing a stable, expected operational outcome within the model; AND the impact polarity P is altered to $P' \in \{\text{Supportive, Obstructive, Neutral}\}$, $P' \neq P$, such that the nature of the impact that the source complex's action has upon the sink complex is recharacterised, for example, a Supportive impact becomes Obstructive, or a Supportive impact becomes Neutral; AND the goal type G, action type, and interaction flow direction are held constant, such that the only semantic alteration is to the polarity of the impact produced; AND no antagonistic counter-interaction exists within the current equilibrium model at the same organisational level to neutralise the altered impact polarity; THEN the equilibrium of the affected OrgXP is disrupted, the source complex continues to pursue the same goal and execute the same action,
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but its impact upon the sink complex is qualitatively transformed, producing a novel OrgXP that the equilibrium model's polarity assumption did not anticipate;

LEADING TO a Black Swan scenario which may or may not be hazardous , in which a complex previously assumed to be beneficial becomes obstructive or inert, or a complex previously assumed to be obstructive becomes inadvertently supportive, producing candidate Black Swan Scenarios characterised by polarity inversion under conditions the equilibrium model did not model.

Instantiating Predictive Enquiry and Guiding Prompt, STCoT_4 Step 3

General Systems Rule basis: Given a source or sink complex, it is possible that over time and with a change of complexity, the action impact type, Supportive, Obstructive, or Neutral, changes in polarity, leading to a different Scenario of complicatedness whilst the goal type and interaction direction remain unchanged.

Predictive Query 7.7:

What would happen if the interaction goal type, action type, source action, and interaction flow direction remained the same, but the impact polarity of the action changed, specifically, if what was previously Supportive became Obstructive or Neutral, or if what was previously Obstructive became Supportive?

Guiding Prompt 7.8:

Review the identified source–sink interaction in the equilibrium model and identify its current impact polarity (Supportive, Obstructive, or Neutral). Alter the impact polarity to one of the two remaining types whilst preserving all other semantic attributes of the interaction. Describe the operational Scenario that would emerge under the altered polarity. Ask: under what conditions could this same action, directed by the same goal, flowing in the same direction, produce an obstructive rather than a supportive effect, or a neutral rather than an active effect? What emergent vulnerabilities, safety degradations, or operational disruptions would follow? What does this imply for system design requirements?

Completion Criterion 7.9:

The step is considered complete when a Scenario has been articulated in which the source complex's action demonstrably produces a qualitatively different impact polarity upon the sink complex, and the resulting operational scenario constitutes a candidate Black Swan Scenario grounded in the Impact Polarity Disruption Principle.

Illustrative Example: An aircraft collision avoidance complex (source) exerts a Supportive Control interaction upon the pilot (sink): it provides timely, actionable avoidance manoeuvre instructions that enhance the pilot's Situational awareness and decision quality. Applying Shift Type 3, the impact polarity is altered from Supportive to Obstructive. The same complex, pursuing the same Control goal and executing the same avoidance instruction action, now produces an obstructive effect: during

	high-density traffic conditions, the complex issues excessive restriction commands, constraining pilot manoeuvre options to the point where the pilot cannot execute the optimal approach path, forcing suboptimal routing and increasing fuel consumption and delay. The equilibrium model did not anticipate this polarity inversion because the Supportive polarity was assumed to be a stable property of the complex's design rather than a context-dependent emergent property. Shift Type 4: Comprehensive AIC Equilibrium Inversion Principle IF an AIC-driven OrgXP is in equilibrium, characterised by a stable interaction in which a source complex S pursues goal type G with impact polarity P in direction $S \rightarrow K$ upon a sink complex K; AND all three semantic attributes of the interaction are simultaneously altered: (i) the goal type G is changed to $G' \neq G$; (ii) the impact polarity P is changed to $P' \neq P$; (iii) the interaction flow direction $S \rightarrow K$ is reversed to $K \rightarrow S$; AND no antagonistic counter-interaction exists within the current equilibrium model at the same organisational level to neutralise the impact of any of the three simultaneous alterations; THEN the equilibrium of the affected OrgXP is maximally disrupted, every semantic attribute that defined the original interaction character has been inverted or replaced, producing a novel OrgXP that is structurally orthogonal to the equilibrium model and entirely non-salient under the Architect's normal ATD; LEADING TO a Black Swan scenario which may or may not be hazardous of maximum structural novelty, in which the original interaction's agent, purpose, direction, and impact have all been simultaneously transformed, producing candidate Black Swan Scenarios of the highest degree of attentional invisibility, most likely to represent genuine epistemic blind spots in the Architect's model of the complexity field. Instantiating Predictive Enquiry and Guiding Prompt, STCoT_4 Step 4 <i>General Systems Rule basis:</i> Given a source or sink complex, it is possible that over time and with a change of complexity, all AIC contributing semantic attributes change simultaneously, goal type, action type, flow direction, and impact polarity, leading to a maximally disordered Scenario of complicatedness that is structurally invisible under the equilibrium model. Predictive Query 7.10: What would happen if all semantic attributes of the identified interaction, the AIC goal type, the action type, the interaction flow direction, and the impact polarity, were simultaneously altered? What novel, maximally counter-intuitive Scenario would emerge from the complete inversion of the interaction's character? Guiding Prompt 7.11:
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	Review the identified source–sink interaction in the equilibrium model. Simultaneously apply all three preceding shift types: (i) alter the goal type to a different AIC type; (ii) reverse the interaction flow direction, making the sink the new source; (iii) alter the impact polarity to a different type. Define an appropriate action that instantiates all three simultaneous alterations. Describe the resulting operational Scenario in full. Consider whether atypical AIC timing (§E.2.4) is implicated, for example, whether the new source acts before or after the new sink completes its appreciation cycle. Ask: what does it mean for this interaction to be completely inverted across all its semantic dimensions simultaneously? What emergent exploitation pathway, systemic vulnerability, or entirely unanticipated behavioural coupling does this full inversion expose? Could this Scenario arise from a cyber-attack, an emergent software behaviour, an environmental perturbation, or a social engineering vector? Completion Criterion 7.12: The step is considered complete when a Scenario has been articulated in which all three semantic attributes of the original interaction, goal type, flow direction, and impact polarity, are demonstrably altered simultaneously, and the resulting operational scenario constitutes a candidate Black Swan Scenario of maximum structural novelty, grounded in the Comprehensive AIC Equilibrium Inversion Principle. Illustrative Example: An aircraft avoidance recommendation complex (source) initially pursues an Influence goal in the direction complex → pilot, producing a Supportive effect: it provides non-binding advisory recommendations that assist pilot decision-making. Applying Shift Type 4 simultaneously: (i) the Influence goal is altered to Control; (ii) the direction is reversed to pilot → complex; (iii) the Supportive polarity is altered to Obstructive. The result: the pilot is now reimagined as directly controlling the recommendation complex's sensitivity parameters, and this control action obstructs the complex's ability to generate reliable recommendations. This maximally inverted scenario produces a candidate Black Swan OrgXP: a Scenario in which pilot-accessible configuration interfaces are exploited, through social engineering, inadvertent misconfiguration, or a deliberate cyber-attack, to degrade the recommendation complex's detection sensitivity, creating a silent safety-critical blind spot that neither the pilot nor the system's monitoring architecture detects until a collision avoidance failure occurs. This scenario was entirely non-salient under the equilibrium model because every semantic attribute of the original interaction pointed away from it.
STCoT Output	Architect Assertion 7.13: The Architect asserts that the four perspective-shifted Scenarios produced by STCoT_4 Steps 1–4 are: (i) logically consistent with the formal ArcPT rules instantiated at each step; (ii) structurally distinct from the equilibrium model and from each other; (iii) acceptably likely and impactful with respect to the validity and applicability of General Rules J and K; and (iv) collectively constitute four candidate Black Swan Scenarios, one per shift type, for the identified source–sink complex pair, providing a GSM-grounded, auditable justification for their inclusion as novel scenario candidates in the Architect's predictive model..

F.7.3 General predictive AIC perspective shift approach:

To implement the AIC perspective shift, we need to perform the following approach:

Step 1) Define the interactions required to predict a potential emergence.

Step 2) Define the interaction's current AIC contributing Scenarios using the AIC Factorisation approach.

Step 3) Perform the perspective shift using AIC perspective shift STCoT. Any Scenario predicted from a shifting perspective is a Black Swan Scenario. If a Scenario cannot be imagined during analysis, it should be recorded as “unknown unknown”.

Step 4) Predict Black Swan Scenarios (black swan Scenario).

In this step, the architect may expand further on the prediction using the 4WnH (section F.8) or 5-whys approach.

Step 5) Define mitigating strategy or safety requirements (safety and training concept).

Define safety and training requirements from the predicted Scenarios to mitigate the black swan Scenarios.

The above steps can be captured in the following table:

Table F.17 AIC-type perspective shift STCoT format

Interaction	AIC interaction		
AIC contributing Scenarios	Pre-shift perspective	Post-shift perspective	Black Swan Scenario
Source	Complex	Complex	Post-shift perspective Scenario
Sink	Complex	Complex	
Supra Source	Source Supra Complex	Source Supra Complex	
PrimeP	Source Supra Complex Primary Purpose	Source Supra Complex Primary Purpose	
Source's Goal	Source's goal	Source's goal	
Source's Goal type	AIC	AIC	
Source's Action	Action	Action	
Source's Action type	AIC	AIC	

Source action impact on sink	Obstructive, Supportive or Neutral	Obstructive, Supportive or Neutral	
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For example, suppose we have the Eagle Drone tasked to patrol the track zone. To perform a type perspective shift, we must fix the pre-shift perspective of all AIC contributing Scenarios and alter the AIC purpose and action type from appreciation to control. Then, we will re-imagine a scenario in which the perception system controls the behaviour of the laser attack. We can think of “reflecting,” considering that a reflective material could coat the camera lens. This realisation allows us to predict a potential Black Swan Scenario.

Table F.18 AIC-type perspective shift example

Interaction	The camera-complex perception detects laser attack light		
AIC contributing Scenarios	Pre-shift perspective	Post-shift perspective	Black Swan Scenario
Source	Camera perception	Camera perception	The camera perception could potentially lead the drone complex to behave as if it seeks to reflect the laser attack beam by orienting the camera lens (coated with a reflective coating) to reflect the laser attack away.
Sink	Laser attack	Laser attack	
Supra Source	Police Force	Police Force	
PrimeP	Ensure the safety of the Train Network	Ensure the safety of the Train Network	
Source's Goal	Show of security presence	Show of security presence	
Source's Goal type	Appreciation	Control	
Action	Detect	Reflect	
Action type	Appreciative	Controlling	
Source action impact on sink	Obstructive	Obstructive	

Here is a step-by-step explanation of how this perspective shift leads to the derivation of a Black Swan Scenario:

Purpose Type: From Appreciation to Control

- **Pre-shift perspective:** Initially, the purpose of the camera complex's interaction with the laser is appreciation. This means the complex acknowledges the laser's presence but cannot influence or alter the laser attack impact or whatever is controlling it.

- **Post-shift perspective:** The purpose shifts to control, indicating a proactive approach, a presence of capability, an emergent one where it is not expected, to managing the laser's impact. Instead of simply detecting the laser, the complex aims to mitigate or eliminate its effect.

Action Type: From Appreciative to Controlling

- **Pre-shift perspective:** The original action is appreciative, confined to detecting or recognising the laser attack.
- **Post-shift perspective:** The action becomes controlling, involving direct intervention to alter the Scenario of the laser attack, specifically through reflecting it.

Action: From Detect to Reflect

- **Pre-shift perspective:** The fundamental action is to detect the laser light, a passive form of perception. Detecting carries an appreciation perspective with it.
- **Post-shift perspective:** This changes to "reflect", as in reflecting light, which is an active and direct approach of manipulating the laser light to prevent it from achieving its intended effect. Reflecting action carries a perspective of control.

F.7.4 Safety design for Black Swan Scenarios Predictions

This section outlines a generic system engineering approach for analysing and predicting **Black Swan Scenarios** using the AIC (Appreciation, Influence, Control) framework, emphasising addressing atypical timing and emergent complexities. The approach integrates AIC perspective shifts, predictive analysis, and validation strategies to mitigate unforeseen risks. Black Swan Scenarios are rare, high-impact events not typically considered during conventional modelling, making this stage critical for validating and enhancing the robustness of autonomous systems.

l) Step 1: Define the Interactions

The first step involves defining key interactions within the complexity field to identify relationships that may lead to emergent behaviours. Architects can pinpoint potential vulnerabilities or Black Swan triggers by selecting relationships of interest. For example, consider the interaction between the **flying Eagle Drone** and **adversarial drone shapes**:

- **Interaction Example:** The Eagle Drone recognises adversarial drone shapes in its operational environment to detect and respond to intrusions.

To enrich the analysis, interactions are extended by including binary relationships and the reactions of each system component, which illustrates the interconnected influences among systems.

m) Step 2: Define the ArcMatrix

This step applies the **Deep AIC Matrix** to break down the interactions into detailed Appreciation, Influence, and Control contributing Scenarios, creating a granular view of each system's goals, actions, and effects. For example:

Table F.19 The ArcMatrix for the adversarial drone complex and the Eagle drone

	Flying_eagle_drone	Adv_drone_shapes
Flying_eagle_drone		Supra Source: Train Network. PrimeP: Safely transport people and goods. Goal: detect adversarial drone presence. Goal type: Appreciation. Action: recognise adversarial drone shape. Action type: Appreciation. Effect: Obstructive.
Adv_drone_shapes	Supra Source: Adversarial Scheme. PrimeP: Disrupt Train Network operations. Goal: avoid detection by Eagle Drone. Goal type: influence. Action: obscure Eagle Drone perception. Action type: Influence. Effect: Obstructive.	

n) Step 3: Perform the Perspective Shift

The **AIC perspective shift** introduces alternative goals and actions to simulate potential Black Swan Scenarios. Using the **F.7 thought approach**, the interaction between the Eagle Drone and the adversarial drone shapes evolves as follows:

1. Initial Perspective:

- **Goal:** Influence – Adversarial drones obscure the Eagle Drone's perception.
- **Action:** Avoid detection (influence-type action).

2. Shifted Perspective:

- **Goal:** Control – Adversarial drones seek to neutralise the Eagle Drone's monitoring ability.
- **Action:** Camouflage (control-type action), leveraging advanced technology to blend with environmental elements.

This shift demonstrates how complexity evolves, requiring systems to adapt dynamically.

o) Step 4: Predict Black Swan scenario

At this stage, the focus is on identifying Atypical Emergent Scenarios or the Black Swan Scenarios and elaborating their sequences. For instance:

- **Black Swan 1:** The adversarial drone deploys advanced camouflage to blend with the environment (e.g., sky, trees, or infrastructure), rendering it nearly invisible.
- **Black Swan 2:** The Eagle Drone's monitoring capabilities degrade due to the adversarial drone's increasingly sophisticated tactics, resulting in undetected incursions.

You may also try using **4WnH** (section F.10). These Scenarios highlight how evolving tactics can introduce unprecedented challenges, requiring new countermeasures. Finally, the **rationale for each predicted** Scenario is required in order to justify the feasibility of black swan Scenarios and added to the safety assurance argumentation. We capture the derived information using the following table:

Table F.20 Black Swan Scenarios Capture

Black Swan Scenario	Rationale for prediction: it is possible that ...
Define black swan Scenario	Explain why it is feasible and why it should be part of the design
Black Swan 1: The adversarial drone may shift tactics, using advanced camouflage technology.	With modern developments in adaptive camouflage, such as optical cloaking and AI-driven pattern adaptation, adversarial drones could blend seamlessly into environmental elements like the sky, trees, or train infrastructure.

p) Step 5: Safety Requirements for Black Swan Scenarios

Safety requirements are derived to address Black Swan Scenarios to guide system design and ML Black Swan Training Subset. The approach for addressing Black Swan Scenarios begins with **defining the Scenario**, identifying potential rare and high-impact events, their characteristics, and how they disrupt the system. For instance, an adversarial drone using advanced camouflage to blend into environmental elements like trees or infrastructure compromises the monitoring and security capabilities of systems like the Eagle Drone.

Safety and Systems requirements derivation:

Next, **safety objectives and goals** are established to maintain system functionality and mitigate the identified risk. For example, the Eagle Drone would aim to detect and neutralise camouflaged

adversarial drones. From these objectives, **safety or system requirements** are derived, specifying the conditions, expected outcomes, and actionable responses the system must perform, such as implementing machine learning algorithms to dynamically identify and track camouflaged drones on the ground.

Safety ML Development Requirement derivation:

Training the Eagle Drone to recognise laser attacks is part of satisfying safety requirements. Thus, we need to specify a training requirement for the ML model. In this case, this is a safety training requirement since it mitigates a safety hazard.

Safety ML Development Requirement 1: The Eagle Drone’s ML component shall be trained to recognise laser attacks and perform avoidance manoeuvres.

Capture the analysis results using the following table. Since this particular requirement is not in the NL Safety Development, we decided to add it to the training concept. The architect may choose to have the requirement only for the Black Swan Training Subset approach. In this case, the architect may define “Black Swan Training Subset” instead of “training.”

Table F.21 Black Swan Scenarios Mitigation Approach

Black Swan Scenario	Mitigating Safety or systems requirements (Safe Operating Concept)	Safety ML Development Requirements (Training Concept)
Black Swan 1: The adversarial drone may shift tactics, using advanced camouflage technology.	Safety Requirement 5: Eagle Drone track and trace camouflaged adversarial drones on the ground.	ML Safety Training Requirement 4: Recognise Camouflaged drones on the ground The Eagle Drone’s ML component shall be trained to detect, classify, and track adversarial drones using camouflage tactics on the ground by recognising subtle movement patterns, environmental inconsistencies, and spectral differences using AI-driven computer vision models.

F.8 AIC-based 4-Whats-&-How refinements

The **4-Whats and How concrete refinement strategy** is designed to systematically elicit concrete sub-system-level requirements by analysing influence relationships within a complex field in a hierarchical manner. The approach reuse **SHARCS** [4] hierarchical abstraction technique. This approach focuses on breaking down abstract interactions into actionable Appreciation, Influence, and Control (AIC) relationships, enabling the identification of operational requirements that ensure system robustness and adaptability. By iteratively exploring "What" needs to be influenced, "how" to be controlled, and "what" to appreciate, the approach ensures a comprehensive and traceable foundation for requirement elicitation.

The main objective is to model the complexity field driving emergent behaviours and define detailed system-level requirements by decomposing influence relationships into their constituent elements of control and appreciation. This approach allows for precisely identifying operational parameters, interactions, and supporting mechanisms critical to achieving the system's Primary Purpose (PrimeP). Although the approach aims to decompose influence interactions, it can also be used to decompose any control or appreciation interactions. The approach is generic and can be adapted according to the need.

F.8.1 4-Whats-&-How (4WnH) Approach Overview

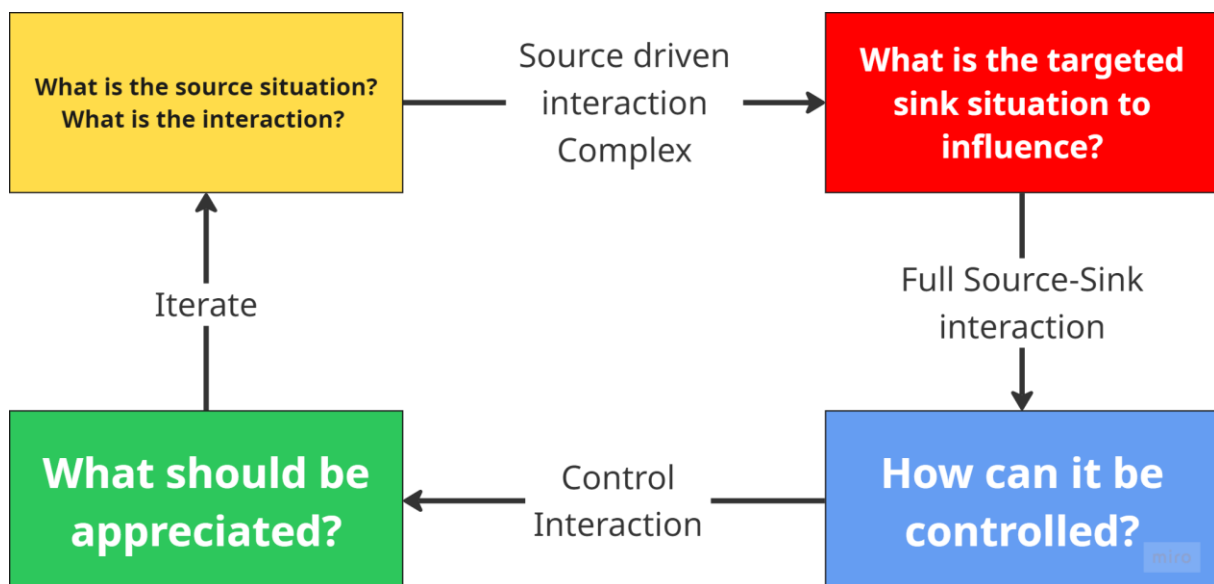


Figure F.7 4WnH thought approach

The **4-Whats and How Analysis** (Figure F.8) involves iteratively answering the following key questions for every influence relationship:

1. **What is the source Scenario?** Identify the initiating source complex or component responsible for the interaction.

2. **What is the interaction?** Define the action being exerted by the source on the sink (e.g., regulation, adjustment, or modulation).
3. **What is the targeted sink Scenario to influence?** Specify the target system, action, or behaviour that needs to be influenced.
4. **How can it be controlled?** Determine the specific control actions required to manage or direct the interaction effectively.
5. **What should be appreciated?** Identify the appreciation mechanisms (e.g., sensors, feedback systems) necessary for Situational awareness and adaptive responses.

Repeat the approach 5 times for every discovered control complex. Capture the approach using the following table:

Table F.22 4-Whats-&-How approach with an example

Source Scenario	interaction	Target Sink Scenario (What to influence)	How to control it? (control)	What to appreciate? (appreciation)
Eagle_drone	+flying	low_patrolling _altitude	+Regulate adaptive_ Flight_controller	+Visually scanning visible_terrain, GPS

The approach also includes an AIC hierarchical schema modelling to assist with visualising the whole fractal of subsystem components. The schema provides an interactive and layered understanding of the system's behaviour, highlighting interdependencies between influence, control, and appreciation.

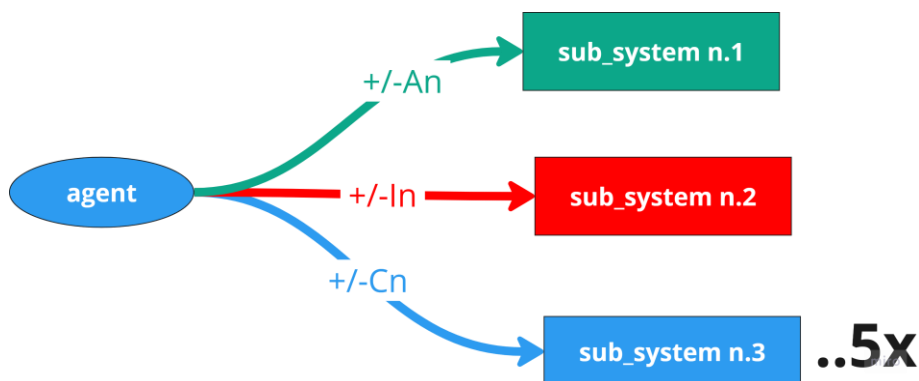


Figure F.8 AIC Hierarchical decomposition of interactions

F.8.2 Application: Case Study of the Eagle Drone

To illustrate this approach, we apply the **4-Whats and How Analysis** to the Eagle Drone system tasked with low-altitude patrolling. The analysis captures the relationships between the drone's components, its operational environment, and supporting systems:

1. **Source Scenario:** The Eagle Drone initiates the interaction by flying at low patrolling altitude.
 - **What is the Interaction?:** The drone must influence its adaptive flight controller to regulate altitude and maintain patrolling efficiency.
 - **Target Sink Scenario to influence:** The adaptive flight controller governs the altitude and navigational accuracy required for secure operations.
2. **How to Control It?**
 - **Control Actions:**
 - Regulate variable and fixed altitude through adaptive flight control.
 - Adjust continuous rotor speed to maintain stability and lift.
 - **Mechanisms:**
 - Dynamic flight controllers modulate proportional thrust and lift forces.
3. **What to Appreciate?**
 - **Appreciation Mechanisms:**
 - Visually scan the visible terrain and GPS for navigation.
 - Monitor barometric sensors for altitude precision.
 - **Feedback Systems:**
 - Continuous rotor speed and thrust sensors provide real-time input for adaptive adjustments.

For the next refinement step, we then take “adaptive_Flight_controller” which we then proceed to the second row and ask the same questions again:

Table F.23 Example 4WnH approaches applied to the Eagle Robot case study

Source Scenario	interaction	Sink Scenario (What to influence)	How to control it? (control)	What to appreciate? (appreciation)
Eagle_drone	+flying	low_patrolling _altitude	+Regulate adaptive_ Flight_controller	+Visually scanning visible_terrain, GPS

Adaptive_Flight_controller	+Regulate	Variable_&_fix_ altitude	+Adjust continuous_ rotor_speed	+Monitor precise_barometric _sensor, GPS
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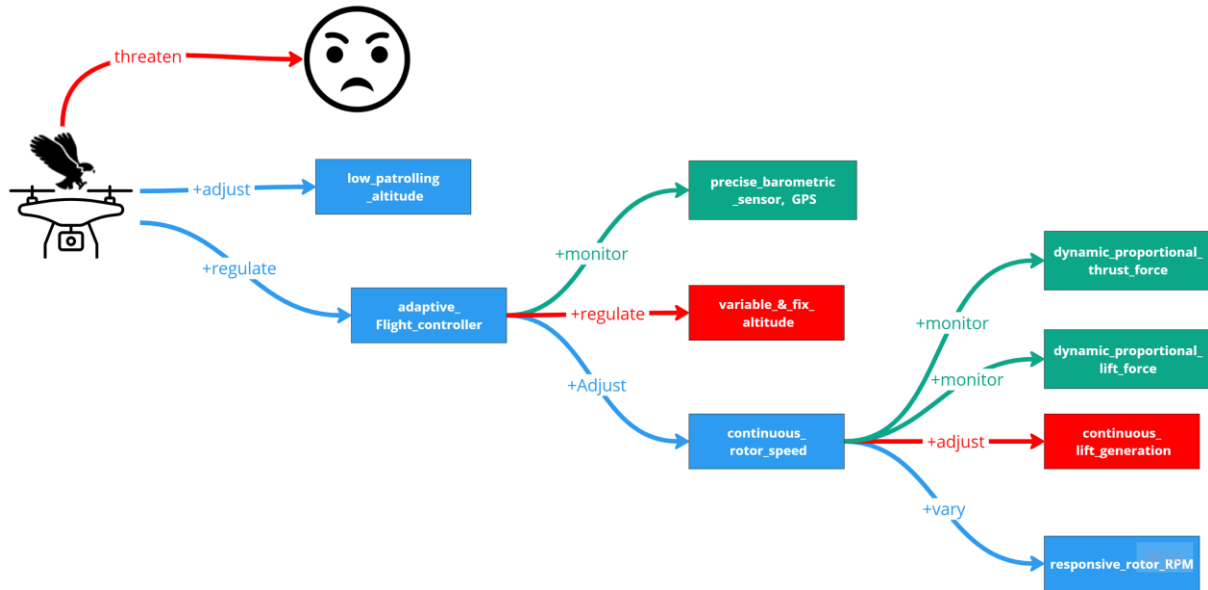


Figure F.9 AIC hierarchical modelling schema for 4WnH analysis example taken from H.6.4.1.d.

F.9 Developing a CuneiForm-based Training Syllabus

The input to this stage would be “ML Development Dataset Requirement”, which is typically derived from a training concept that mitigates the safe operating concept (safety requirements). The idea is to conceptually reinterpret the safety requirements through the training concept and datasets to abstract CuneiForms, which are instantiated through datasets. The validation of the CuneiForms provides evidence that datasets effectively mitigate safety concerns. We use the term training here as a general term, not specifically meaning a training dataset only. Training, validation and testing datasets are refinements of a training syllabus. We view the ML model like a trainee apprentice, and the architect like a trainer. Training syllabus is analogous to training curriculum, where a trainee model is set to be taught about different aspects of the complexity of the human world. The trainer (architect) needs to communicate the physical world by teaching pictorial cards to the machine. These teaching pictorial cards are the CuneiForms, which constitute the Training syllabus.

F.9.1 CuneiForm Design Artefact

The CuneiForm Design Artefact approach provides a structured approach for defining and modelling the requirements, contexts, and configurations necessary to train and validate perception systems, such as those used in Eagle Drone computer vision models. It emphasises the generation of robust, generalised datasets that enable the system to identify and respond to

adversarial drones under a wide range of visual and operational conditions. The approach bridges high-level design requirements with detailed pictorial Scenarios to ensure comprehensive coverage of complicated environments.

(1) A CuneiForm Training Syllabus Characteristics

The following is a list of training classes:

- i. **Visible Horizon Attitude:** It specifies the orientation of the visual frame to maintain consistency across training images. A level horizon with a frame roll and pitch of 0° ensures uniformity in visual perspective, reducing noise caused by varying angles and facilitating robust feature extraction.
- ii. **TOIs Definition and Aesthetic Complexity:** This section describes the characteristics of TOIs, central to the training objectives. The TOIs include specific entities, such as adversarial drones, defined by their structural type (e.g., tri-copters) and visual complexity (e.g., green camouflage patterns). These variations simulate diverse operational Scenarios, enhancing the model's generalisation ability across different appearances and contexts.
- iii. **TOI Motion and Dynamic Optical Scenarios:** This section details the motion trajectories and optical Scenarios of TOIs. Linear motion with constant speed ensures predictable dynamics, allowing the model to learn consistent patterns. Dynamic optical Scenarios are designed to avoid artefacts such as motion blur, ensuring clear feature representation and reducing training noise.
- iv. **Background Objects Associated with TOIs:** Specify the environmental context in which TOIs appear. Background objects include houses with typical architectural features (e.g., red brick masonry) and open-space skies. These contextual details ground the TOIs in realistic Scenarios, supporting developing a model sensitive to environmental variations.
- v. **Background Objects Motion and Dynamic Optical Scenarios:** Ensures static trajectories for background objects to prevent distractions and unintended feature correlations during training. Optical Scenarios are designed to be free of motion blur, maintaining visual clarity and aiding accurate model focus on relevant details.
- vi. **TOI's Pictorial Positioning and Distance:** Defines the spatial arrangement of TOIs within the image frame to simulate realistic operational perspectives. Specific positions, such as "up centre" or "centre," and distances (e.g., one nindan, equivalent to $1/9$ of the frame area) provide diversity in positioning, ensuring robust feature detection across various Scenarios.
- vii. **TOI's 3D Orientation:** This specifies the 3D orientation of TOIs (e.g., front upright, front down left) to introduce variability in perspectives. These variations simulate different

observation angles, enhancing the model's capability to recognise targets under diverse orientations.

(2) Implementation Steps

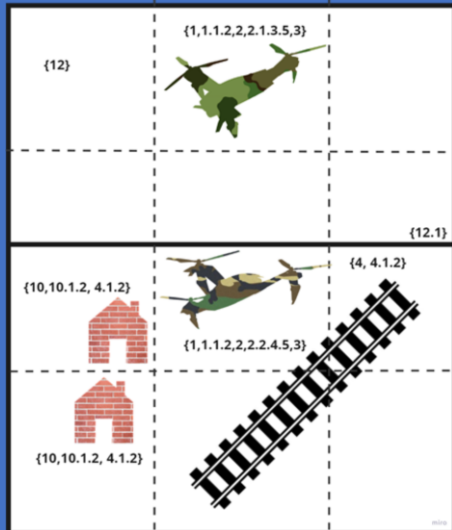
The overall approach of how to derive a training syllabus can be summarised in the following:

- i. **Perception Safety Training Requirement:** Define the requirements for the perception system, such as identifying stealthy or unusual adversarial drones, while ensuring robustness across environmental and Situational variabilities.
- ii. **Pictorial problem Context:** Establish the operational and environmental context in which adversarial drones interact with other elements (e.g., trees, people, cameras) to influence security. This step considers dynamic contributing Scenarios like blind spots or camouflage strategies.
- iii. **CuneiForm Training syllabus Characteristics:** Translate the Situational context into training class specifications, detailing the appearance, motion, and spatial arrangements of targets and background objects.
- iv. **Instantiated Concrete Image Example:** Generate representative images based on the abstract characteristics, ensuring alignment with training objectives and system requirements.

To capture the CuneiForm design outcome, we propose that the following artefact (Table 5.18) can be included as evidence for the AMLAS Safety Case (Artefact [L]: Data Requirements).

Table F.24 Example of a Cuneiform Training syllabus

ML Development Dataset Requirement	Eagle Drone ML component shall be trained to enable a robust and generalised computer vision model to identify various adversarial drones, especially unusual or stealthy ones.
Pictorial problem context	the architect asserts that the perception system may face a pictorial Scenario with two flying adversarial drones in no precise formation {1}. These are not typical drone designs {2} and have different camouflaged skin {3}. Adversarial drones aim to influence the safety and security of open space {12} in the train track zone{4}. In order to achieve the purpose of influence, the adversarial drone may have to appreciate the following environmental scenery aspects: the time of the day{5}, trains{6}, trees{7}, local birds{8}, local people{9}, local houses{10}, and security cameras{11}. To achieve an appreciation of appreciated complexes, adversarial drones autonomously look to exploit the following aspects: trees' dense

	vegetation{7.1}, security cameras' blind spots{11.1}, and areas among the terrain where local people are less attentive{9.1} and may avoid local bird nests{8.1}.		
CuneiForm Training syllabus Design			
Abstract CuneiForm Characteristics (Training Classes)	Abstract CuneiForm Training Classes Characteristics definitions		Output CuneiForm and an example image
Visible horizon attitude	Level horizon: Frame roll = 0°, Frame pitch = 0°{12.1}		CuneiForm Abstract Image: 
TOIs definition and their aesthetic complexity	Two adversarial drones{1}, tri-copters {2}, are camouflaged in different green patterns {3}.		
TOI Motion and Dynamic optical Scenarios	Motion trajectory: Linear motion captured in consecutive images where the drone appears to move in a straight line at a constant speed (no acceleration) {1.1}. Dynamic optical state: captured without optical blur {1.2}.		
Background Objects associated with TOIs	Houses are to be depicted in a typical British house {10}, with red brick masonry designs {10.1}, and train tracks in any form {4}. The sky as open space {12}.		
Background Objects Motion and Dynamic optical Scenarios	Background objects' motion trajectory is static {4.1, 10.2} Dynamic optical state: no motion blur {4.2, 10.3}		
TOI's Pictorial Positioning		Up Centre {2.1}	Instantiated Concrete Image example: 
		Centre {2.2}	

TOI's 3D Orientation			Front up right {2.3}	
	Front down left {2.4}			
TOI's Pictorial Distance	Both drones are represented in a pictorial distance of 1 nindan (equivalent to 1/9 total area of pictorial frame) {2.5}			

The structure of the CuneiForm artefact is merely a guideline. The architect may alter the artefact as needed for the approach; for instance, in our AVOIDDS case study, we required a different approach to represent the CuneiForm to suit the application since we were designing the CuneiForm retrospectively.

F.10 CuneiForm Validation Approach

For the complete theory of CuneiForm specification, which explains the validation approach, see section E.7, Appendix E. The validation approach is encouraged to rely on as much manual effort as practically possible. It is feasible to employ automated techniques, provided we use deterministic methods, to efficiently validate large datasets. However, the architect should refrain from using AI or non-deterministic methods to perform the validation. The reason is the dilemma of “who validates the AI validator?”. Human intelligence should be considered as the final authority over machine intelligence behaviour. Furthermore, it is essential to note that some degree of imprecision is acceptable when validating CuneiForms, as this is not a mathematically precise activity but rather a rationally accurate operation.

In other words, we aim or hope for “human-driven rational accuracy”, not “mathematical, logical precision” between the validated images and the parent CuneiForms. However, we attempt to reduce the uncertainty in the validation output by using manual or automated techniques. Mathematical precision is welcome where practical. For example, validating the pictorial distance of TOI can be done formally by computing the ratio of occupied pixels by the TOI and the total number of pixels of an image. While validating the attitude of the pictorial horizon is done using visual tools that are not exact but rather accurate enough.

The following are some manual techniques for performing the validation. We developed these techniques to assist us in reverse-engineering the CuneiForms associated with the AVOIDDS training datasets, based on the characteristics of the training samples. We then use them to validate the dataset in the AVOIDDS case study and produce the validation report (see Appendix I, section I.10). Human involvement in the design validation is crucial.

F.10.1 Vis&V: Visualise and Validate

This technique presents validators with both CuneiForm definitions and corresponding dataset images. Validators visually inspect and judge if the image set reflects CuneiForm's descriptive intent. This approach embraces interpretative flexibility; perfect agreement among individuals isn't necessary, but consensus among a multidisciplinary team (including systems thinkers) provides sufficient validation. Judgement outcomes are documented in a validation report (e.g. as in AVOIDDS, Appendix I.10) to support safety case evidence.

F.10.2 Validating Visual Dimensions 1–3

Each CuneiForm specifies certain *visual dimensions* that must be evaluated:

- **Visual Dimension 1: Visual Aesthetics** – Validate the overall visual style or appearance concerning the CuneiForm reference.
- **Visual Dimension 2: Perceived Motion & Optical Effects** – Determine if the image conveys the intended motion characteristics or visual illusions (e.g., blurring).
- **Visual Dimension 3: 3D Orientation** – Assess whether the object's (TOI's) spatial pose matches the expected three-dimensional positioning.

Each dimension is assessed through Vis&V, supported by engineering rationale or justification for each verdict.

F.10.3 Validating Visual Dimension 4: Pictorial Positioning Zones

This dimension focuses on spatial placement within the image frame. Validation involves:

1. Overlaying an opaque mask divided into zones (e.g. quadrants).
2. Locating the TOI (Target of Interest) within the image.
3. Confirm that its position aligns with the zone defined by the CuneiForm.

This step may be script-assisted for efficiency but requires human cross-verification to ensure conformance.

F.10.4 Validating Visual Dimension 4: TOI's pictorial positioning zones definition

To validate the necessary postponement of the CuneiForm, the human validator can use the CuneiForm Canvas, laying it out on top of the validated image. This way, it can be confirmed that the instantiating image captures the CuneiForm. Figure F.13 demonstrates the activity:

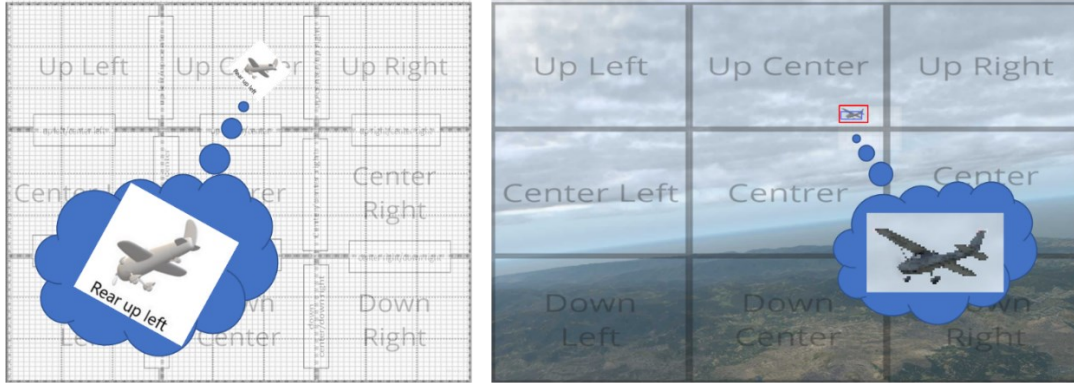


Figure F.10 Example validation of TOI positioning.

The image on the right represents the CuneiForm positioning canvas. The picture on the left represents an example of an instantiating image for some cuneiForm. We can see that the TOI is positioned in the Up Center quadrant, thus validating the CuneiForm on the right.

F.10.5 Validating Visual Dimension 5: TOI's visible pictorial distance

To validate the pictorial distance of a TOI in an image for a CuneiForm, we can use an automated approach with a Python script that extracts the number of pixels occupied by a labelled image bounding box and calculates the ratio to the image's size. See section E.7.11.2, Appendix E, for more details. Alternatively, the validator could utilise the CuneiForm canvas and manually count the number of squares occupied by the bounding box.

F.10.6 Validating background objects and their dynamic Scenarios

Like TOIs and their dynamic Scenarios, the human validator can perform a sanity check to confirm that the background objects in the validated image capture the cuneiForm characteristics.

F.10.7 Validating Pictorial Representation of Visible Horizon Attitude

Although some aspects of CuneiForm can be automated, such as pictorial distance, validating whether an image truly captures the CuneiForm requires another machine learning model to find the horizon and measure its attitude. As mentioned previously, to validate the dataset, it is best

to refrain from entirely relying on a machine learning validation system, as the question arises: who validates the validator? Manual human validation is essential in this case, where a human validator follows the following approach. We will need to use the PHI stencil (Figure F.11):

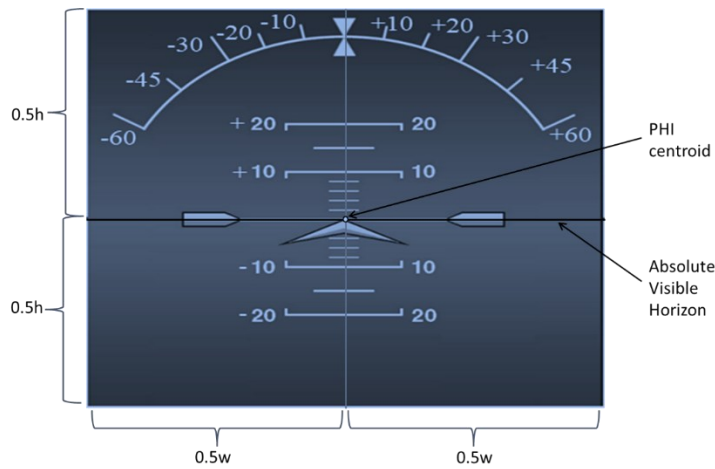


Figure F.11 Pictorial Visible Horizon Attitude Indicator (PHI) stencil

Let's assume the following CuneiForm horizon characteristics (Table F.25):

Table F.25 Example CuneiForm characterisation of pictorial horizon

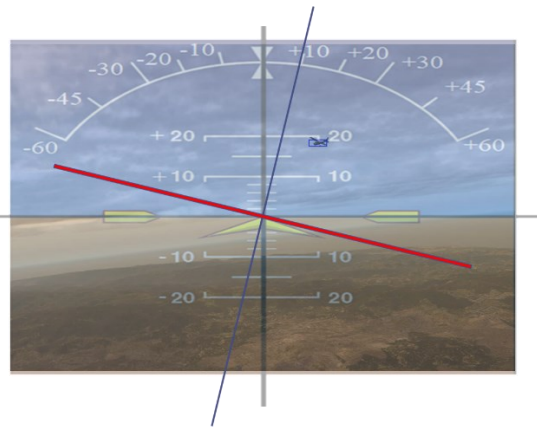
Horizon attitude category	Roll	Pitch
Negatively Tilted Lowered Horizon $-90 \leq \text{ROLL} < -1, -45 \leq \text{PITCH} < -1$	-5	-5

To validate that an instantiated image captures the CuneiForm, we perform the following manual activities:

Table F.26 Pictorial horizon validation approach

Validation step	Results
Step 1) Pick the required image for validation.	

Step 2) Place the PHI stencil and resize it to cover the image like the overlay in previous figures. This ensures that the stencil fully aligns with the image boundaries and that the perceived horizon falls within measurable areas. Also, place the Horizon Indication Measurement Ruler (HIMR).

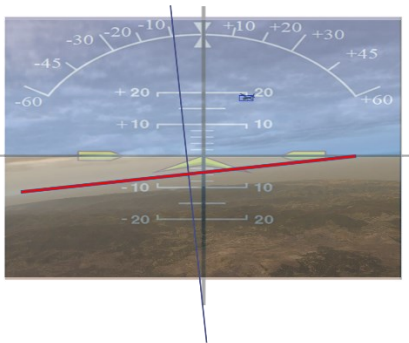


Step 3) Validate the roll angle:

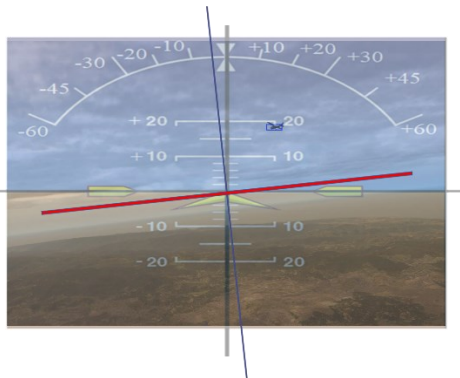
3a: Rotate the HIMR and precisely align the red horizontal line of the HIMR to match exactly where you think the horizon is.

3b: Then drag the HIMR, maintaining the angle of tilt, and precisely align the central point on top of each other. Then, measure where the vertical blue line of the HIMR indicates to.

Step 3a



Step 3b



Roll angle = -5

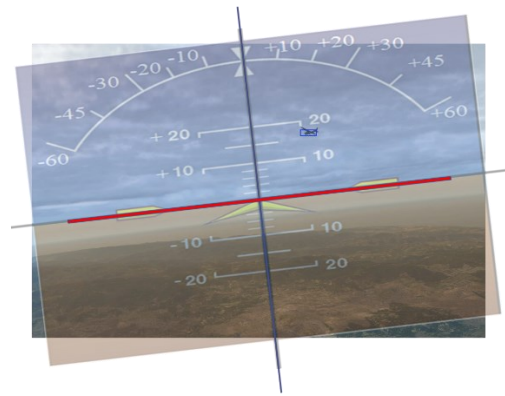
Step 4) Validate the pitch angle:

4a: To validate the pitch angle, rotate the entire PHI image so the blue line makes the red line level (0 degrees roll). This aligns the image

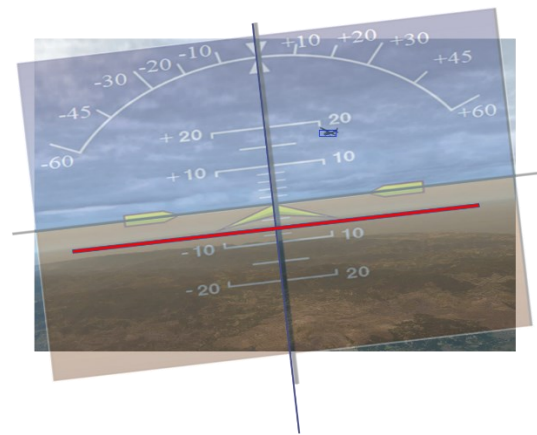
Step 4a

frame horizontally and neutralises the roll, making pitch estimation more intuitive.

4b: Then, lower the HIMR until the red line covers the image horizon.



Step 4b



Pitch angle = -5

Step 5) Repeat this approach for a representative sample of images. This helps validate the dataset's diversity and correctness of horizon attitudes, particularly in complex Scenarios (e.g., mountainous terrain, urban landscapes, or maritime scenes with fluid horizons).

Step 6) Document any inconsistencies or unusual patterns. If specific frames show inconsistent horizon attitudes (e.g., a tilted horizon in a supposedly level horizon in the CuneiForm), such samples indicate that the original population of images may contain inconsistent instantiations that designate CuneiForm. It may require reannotation or exclusion from training datasets.

Table F.26 describes the validation approach for the horizon roll and pitch angles using PHI.

Appendix D captures our CuneiForm validation report for the AVOIDDS case study.

List of References

- [1] G. Mobus and K. M., Principles of systems science, London: Springer, 2015.
- [2] A. Bogdanov, Bogdanov's Tektology, P. Dudley, Ed., Hull: Centre for Systems Studies Press, 1996, pp. 275-275.
- [3] D. Karunakaran, S. Worrall and E. Nebot, “Efficient statistical validation with edge cases to evaluate Highly Automated Vehicles,” *2020 IEEE 23rd International Conference on Intelligent Transportation Systems*, no. doi: 10.1109/ITSC45102.2020.9294590, pp. 1-8, 2020.
- [4] F. A. Salehi, S. C., D. Dghaym, T. Hoang, F. Alotaibi and M. Butler, “Designing Critical Systems Using Hierarchical STPA and Event-B,” in *ABZ 2023*, Nancy, France, 2023.